

BFS Traversal

Step 1: Start at E

Legend

Complete

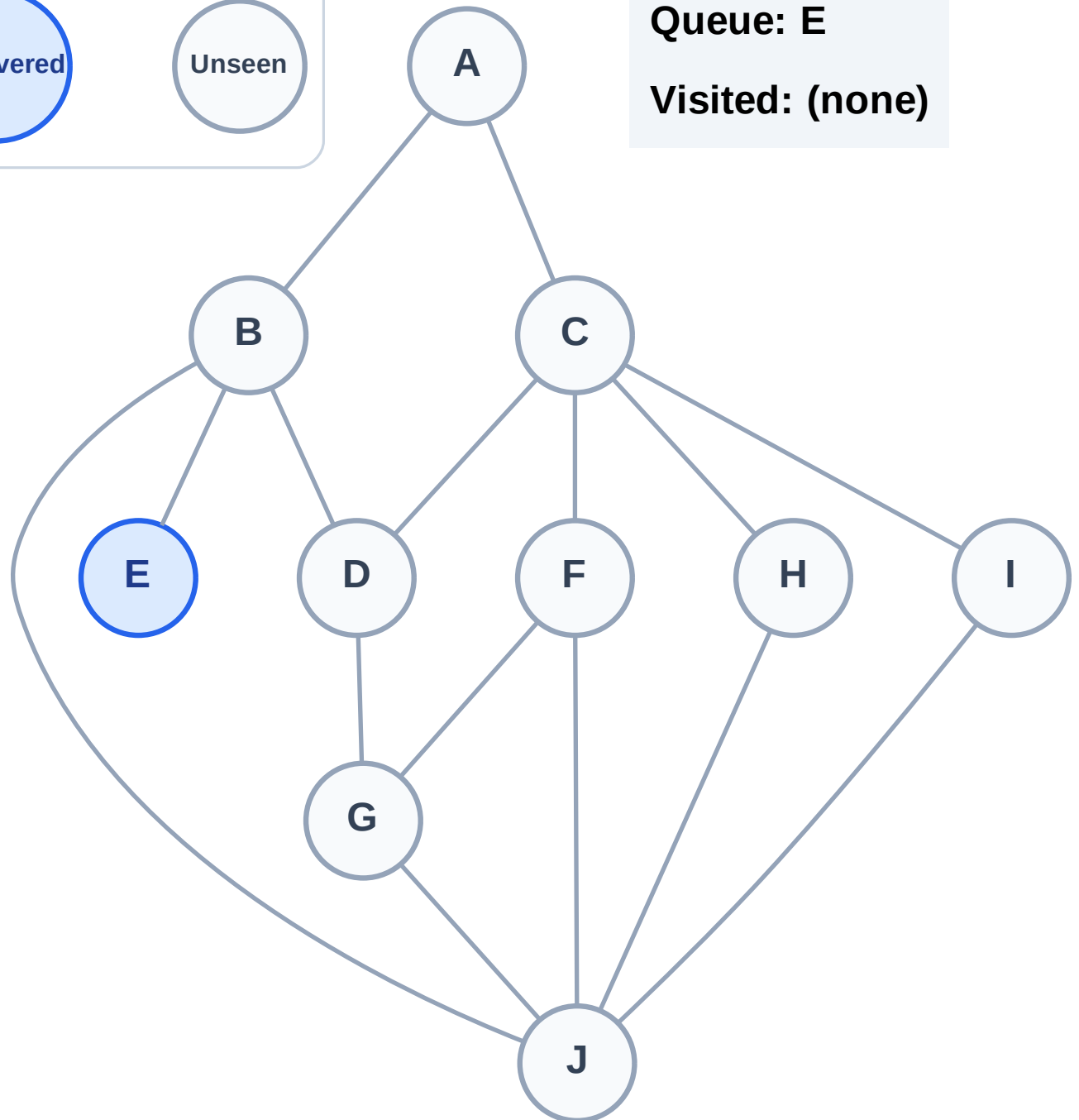
Processing

Discovered

Unseen

Queue: E

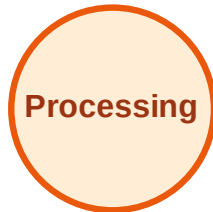
Visited: (none)



BFS Traversal

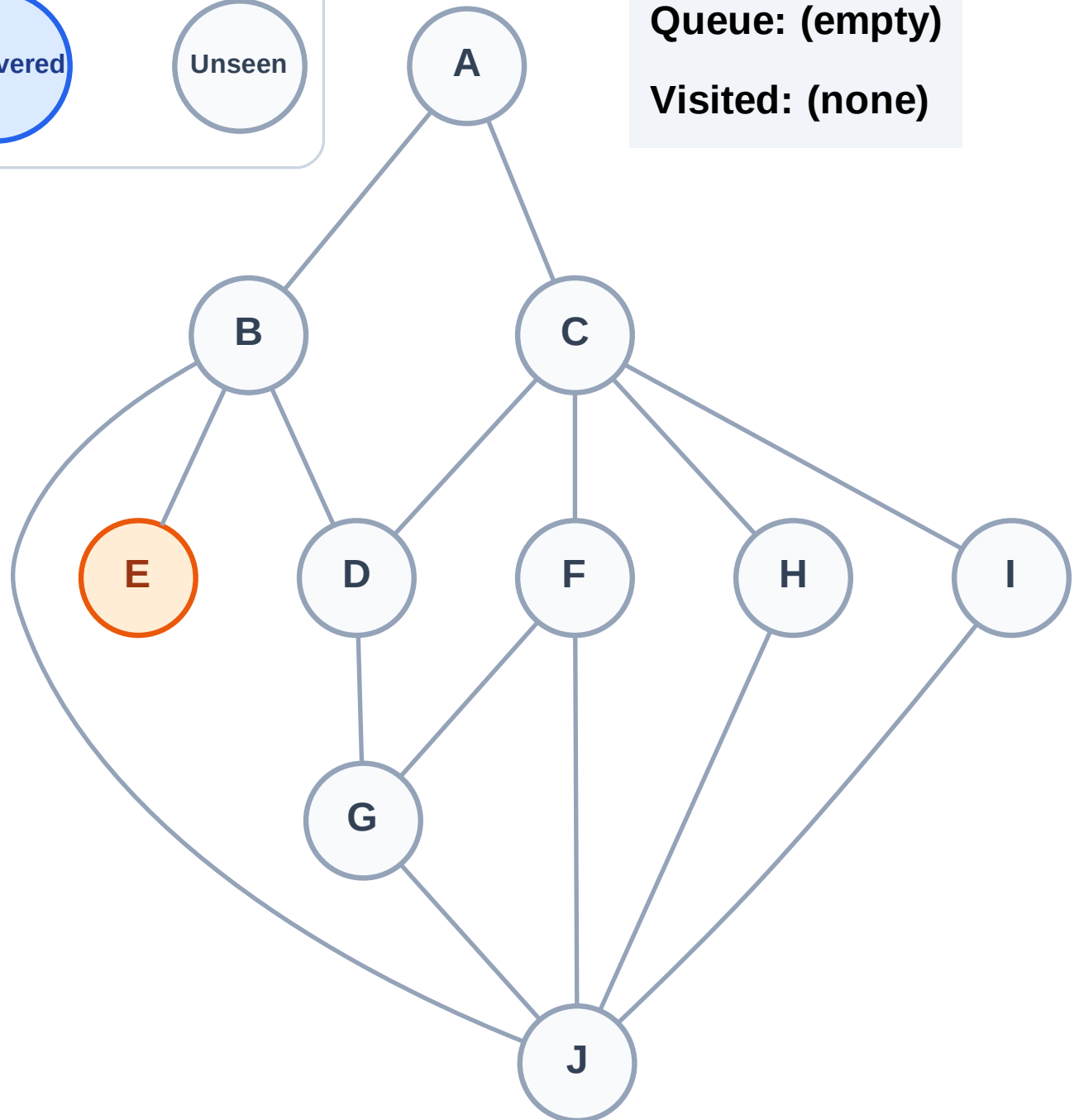
Step 2: Process node E

Legend



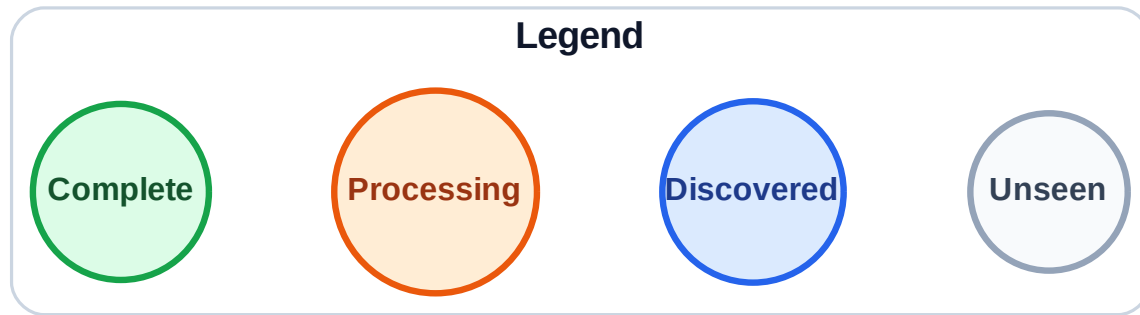
Queue: (empty)

Visited: (none)

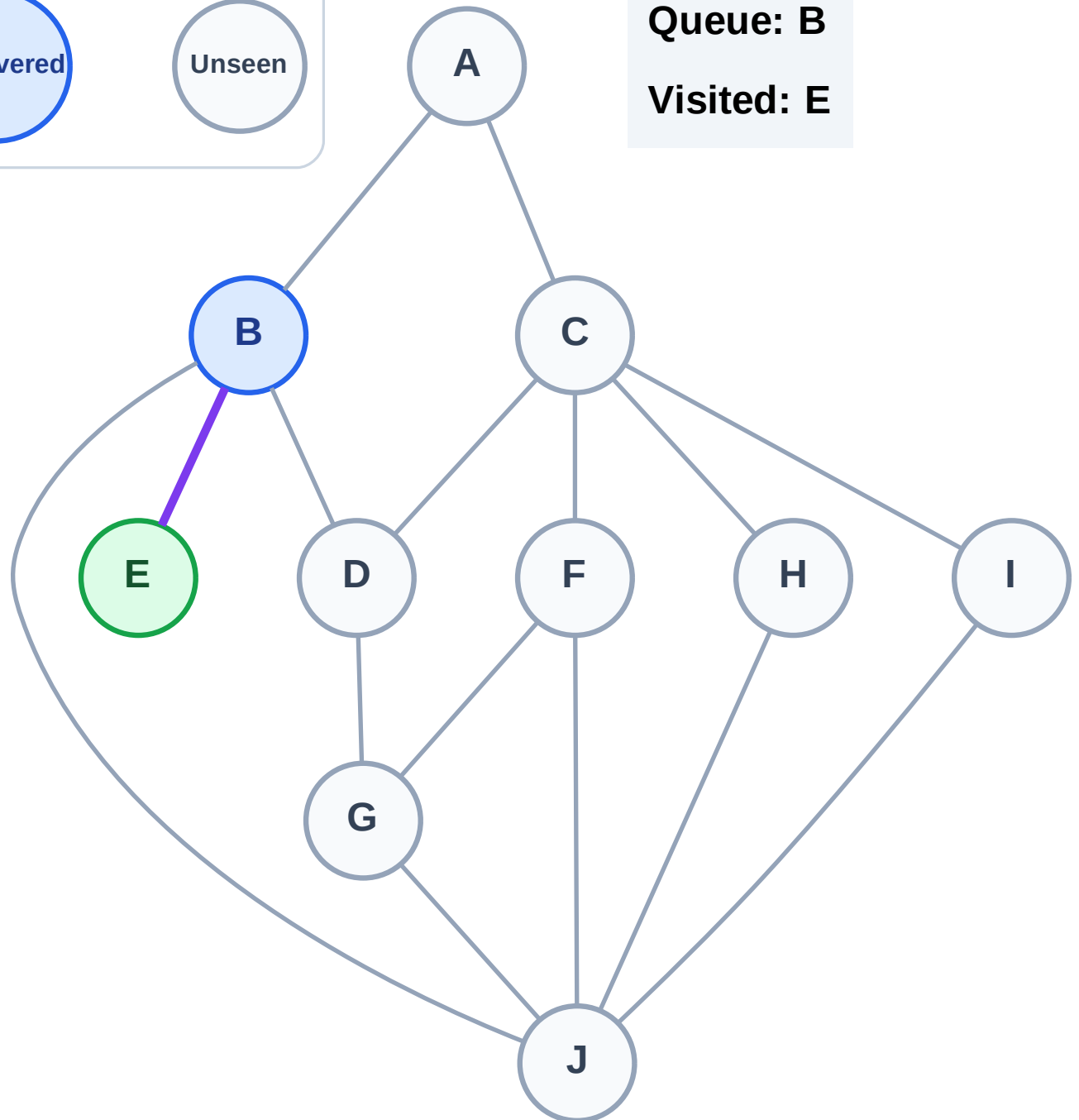


BFS Traversal

Step 3: Discover B from E



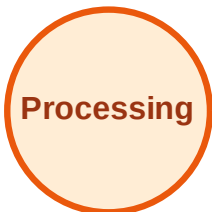
Queue: B
Visited: E



BFS Traversal

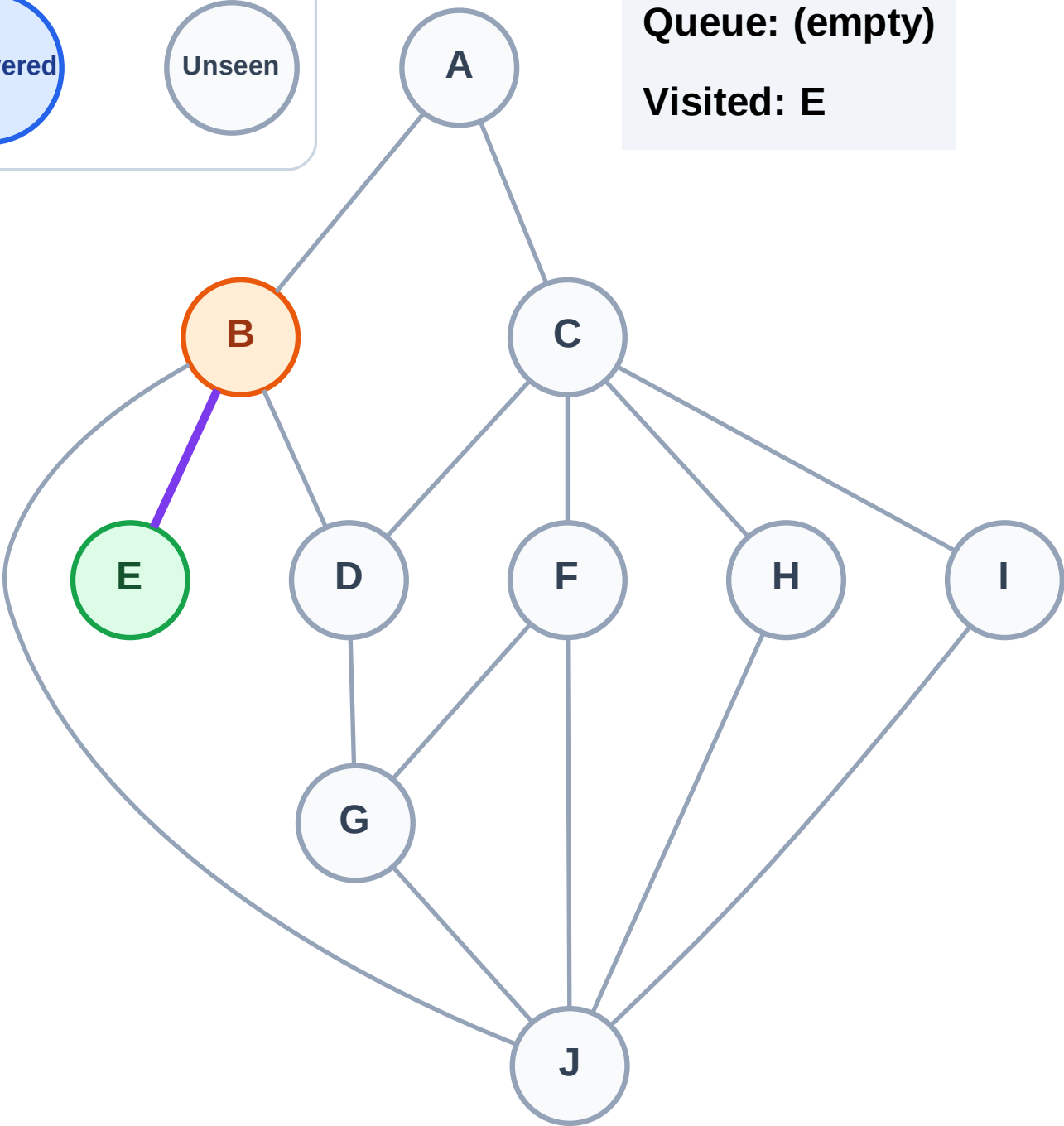
Step 4: Process node B

Legend



Queue: (empty)

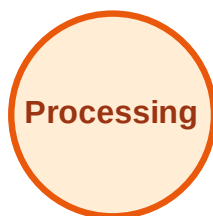
Visited: E



BFS Traversal

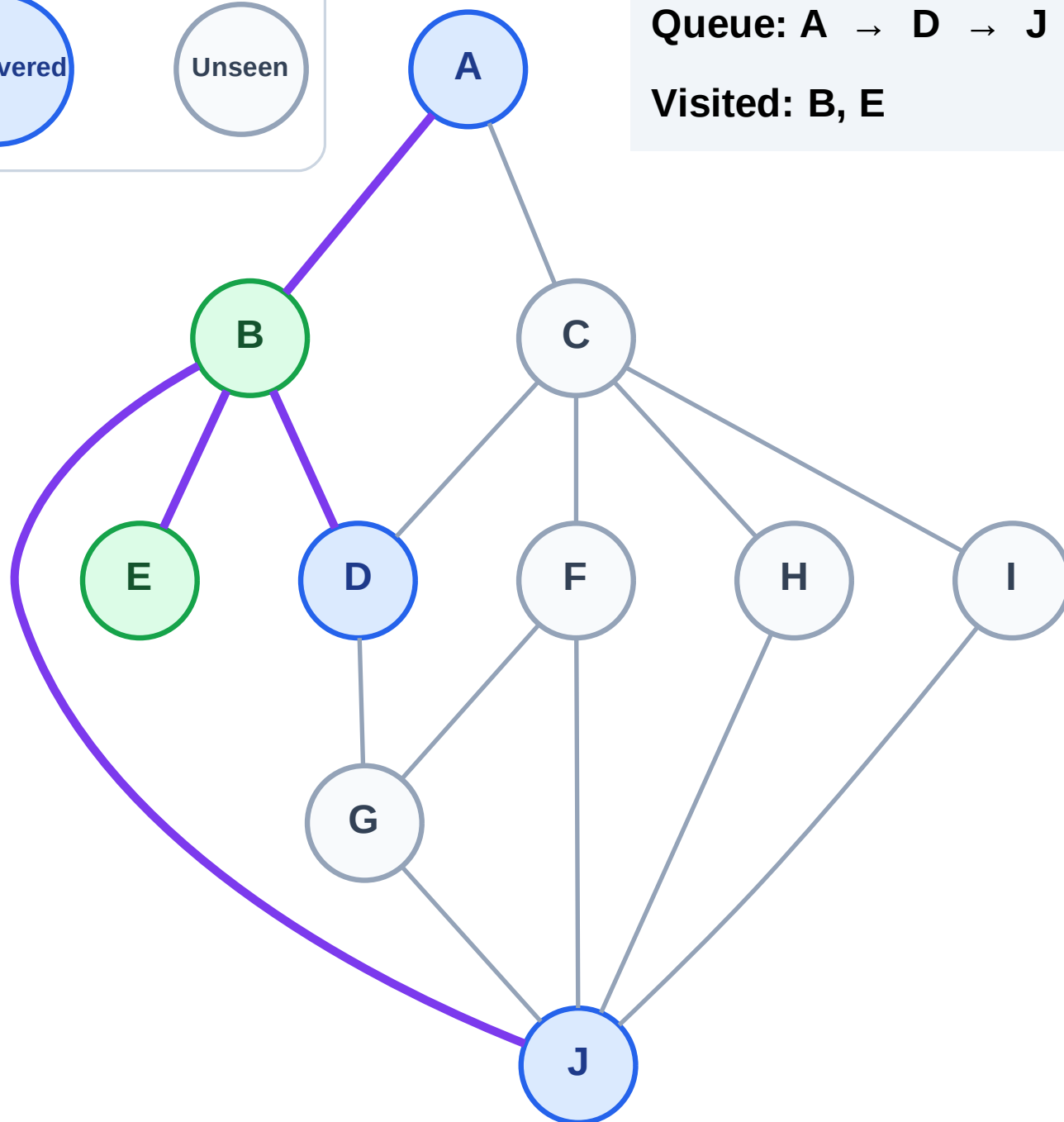
Step 5: Discover A, D, J from B

Legend



Queue: A → D → J

Visited: B, E



BFS Traversal

Step 6: Process node A

Legend

Complete

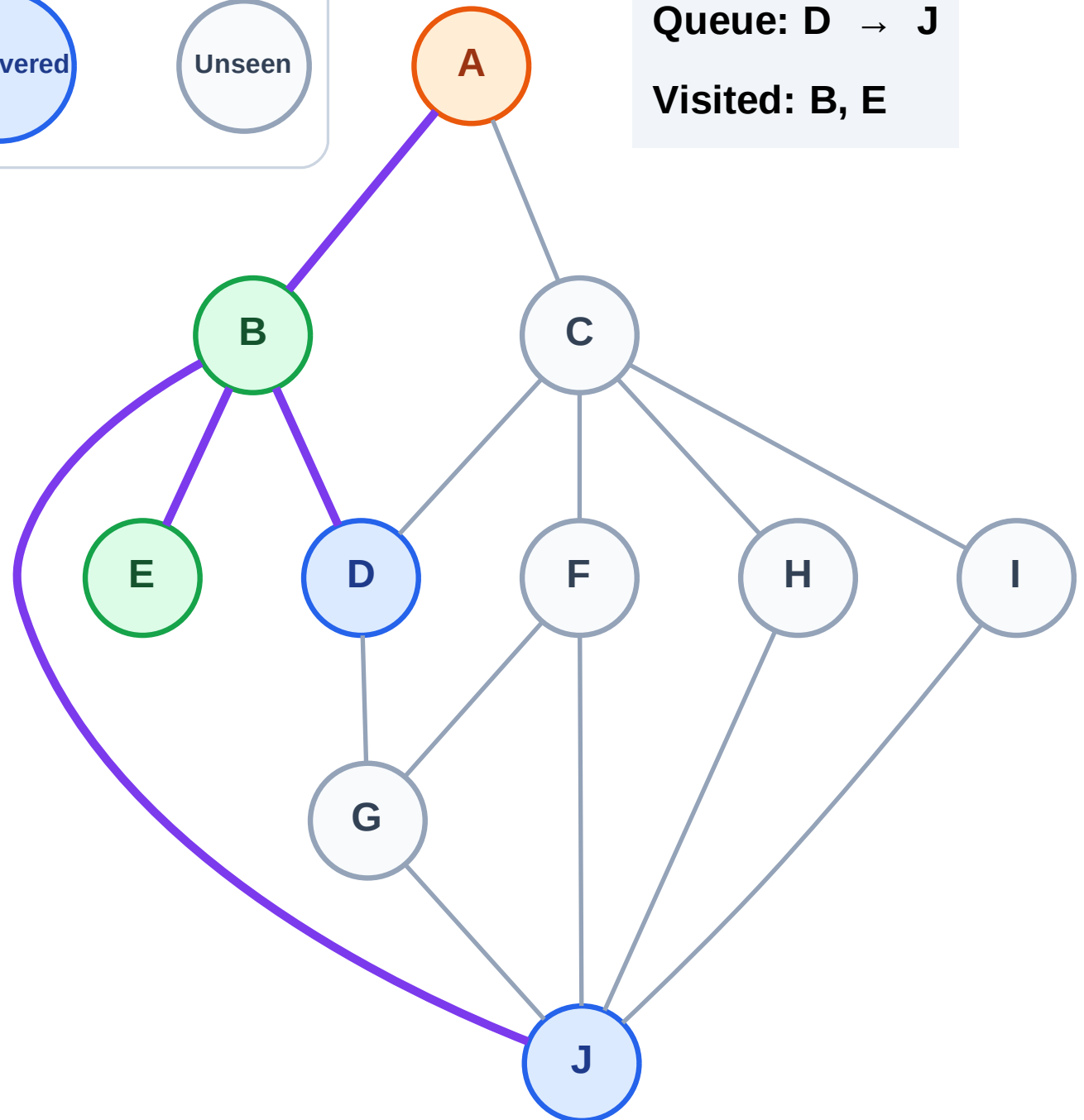
Processing

Discovered

Unseen

Queue: D → J

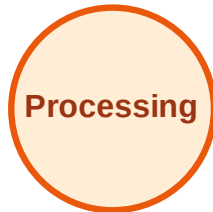
Visited: B, E



BFS Traversal

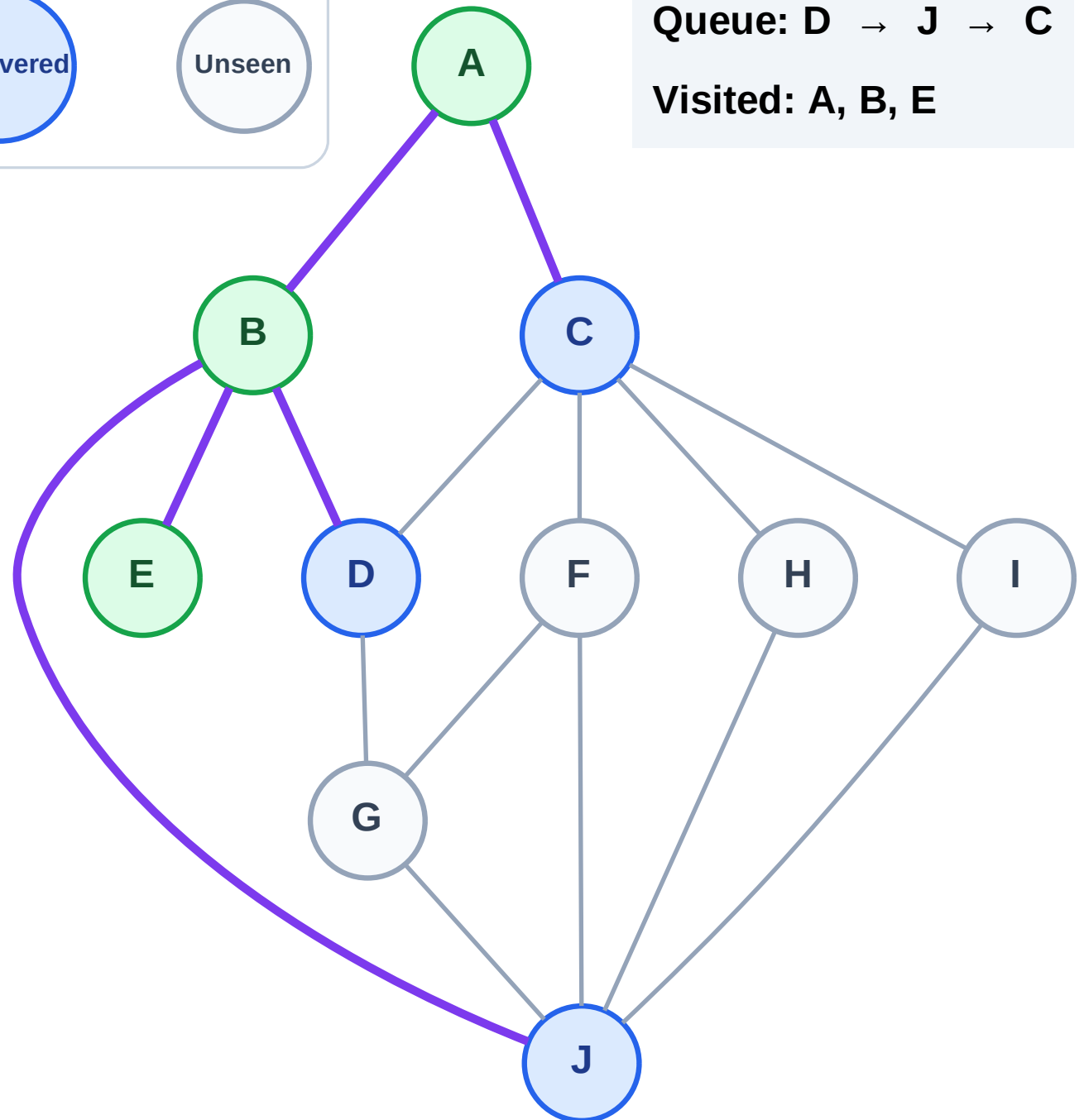
Step 7: Discover C from A

Legend



Queue: D → J → C

Visited: A, B, E



BFS Traversal

Step 8: Process node D

Legend

Complete

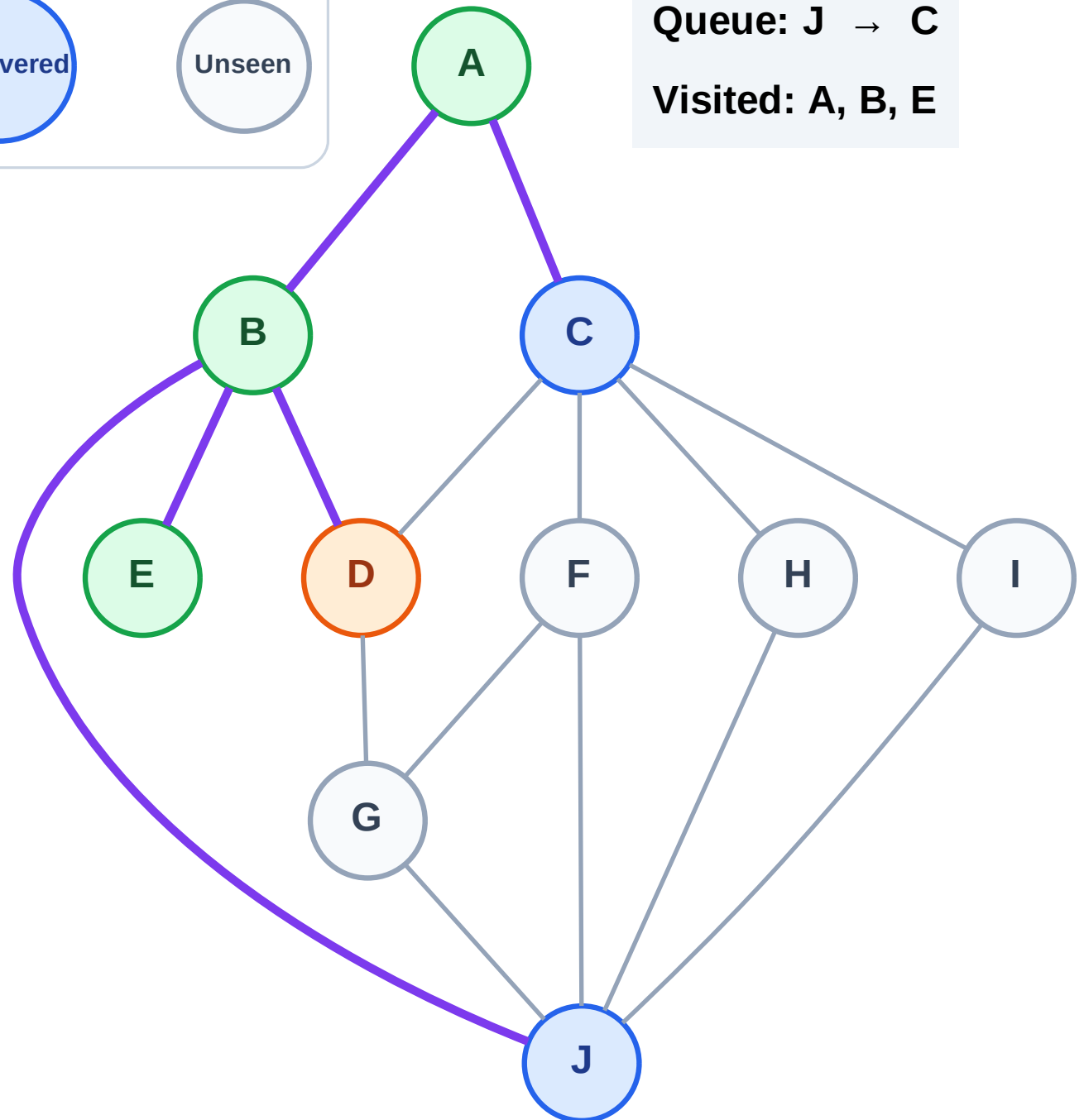
Processing

Discovered

Unseen

Queue: J → C

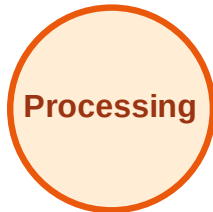
Visited: A, B, E



BFS Traversal

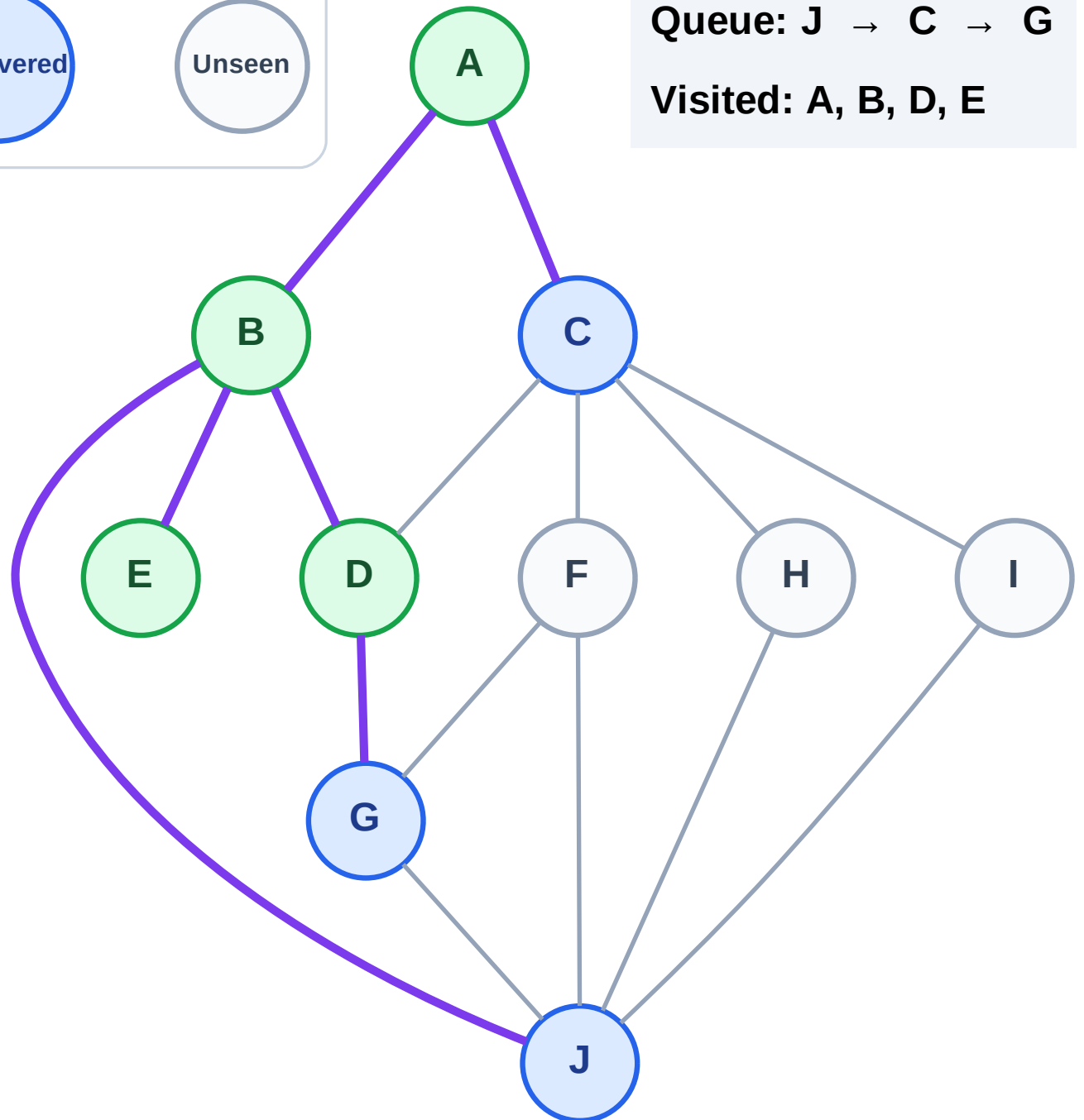
Step 9: Discover G from D

Legend



Queue: J → C → G

Visited: A, B, D, E



BFS Traversal

Step 10: Process node J

Legend

Complete

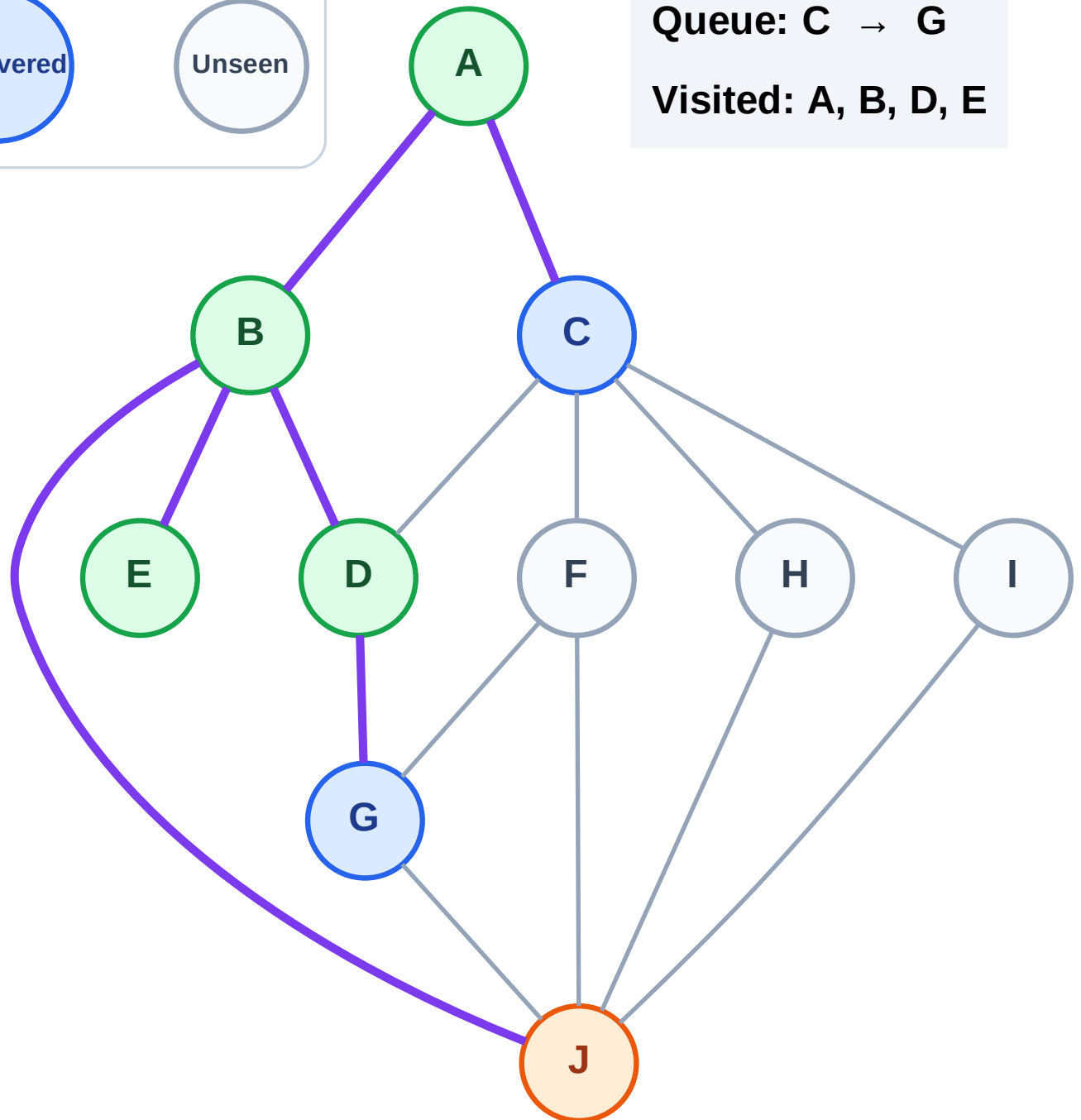
Processing

Discovered

Unseen

Queue: C → G

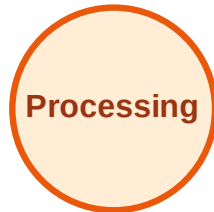
Visited: A, B, D, E



BFS Traversal

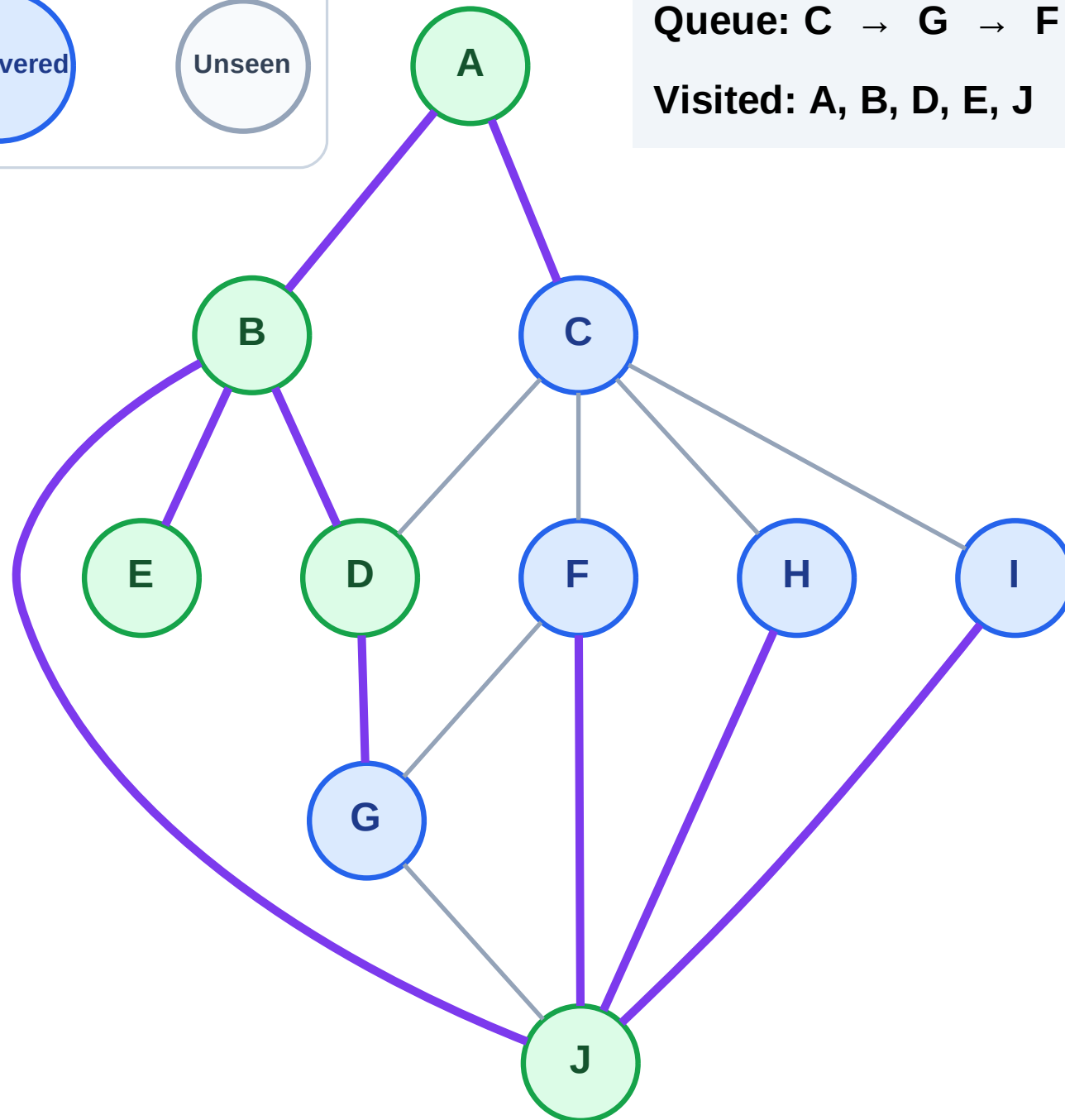
Step 11: Discover F, H, I from J

Legend



Queue: C → G → F → H → I

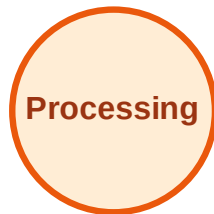
Visited: A, B, D, E, J



BFS Traversal

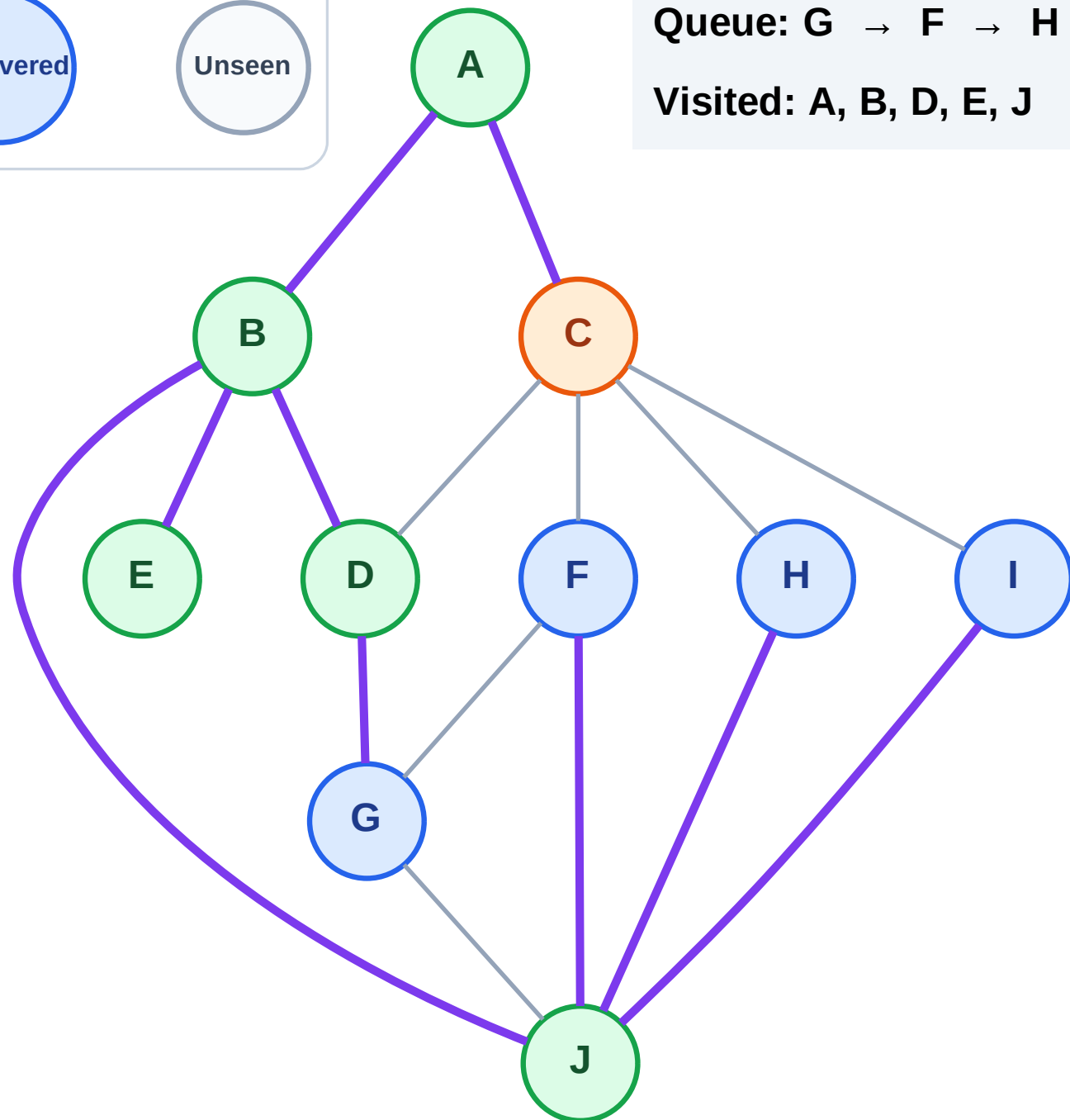
Step 12: Process node C

Legend



Queue: G → F → H → I

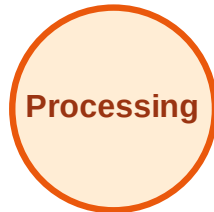
Visited: A, B, D, E, J



BFS Traversal

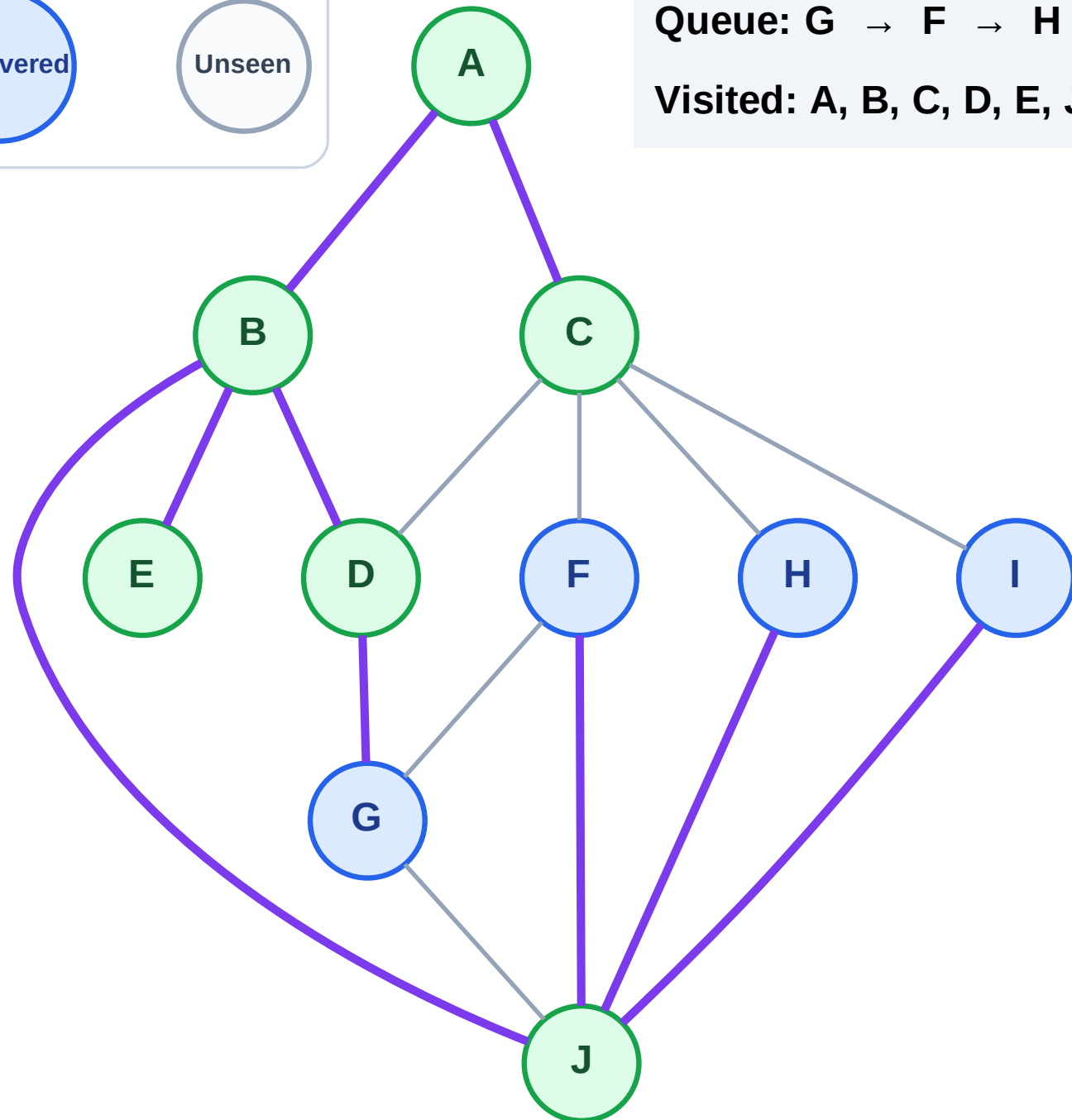
Step 13: No new neighbors from C

Legend



Queue: G → F → H → I

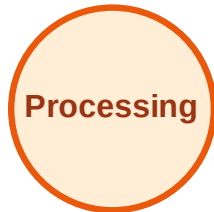
Visited: A, B, C, D, E, J



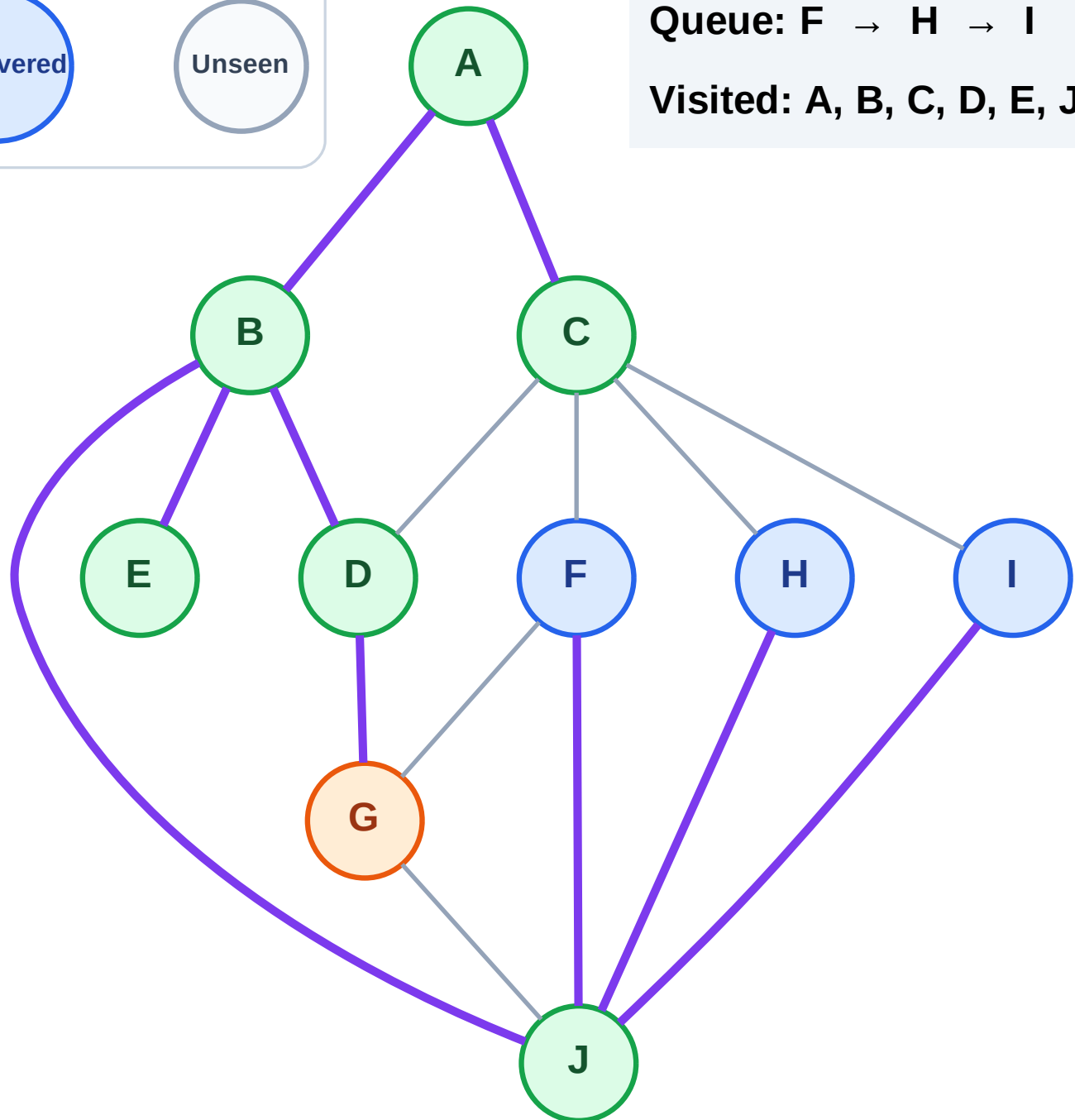
BFS Traversal

Step 14: Process node G

Legend

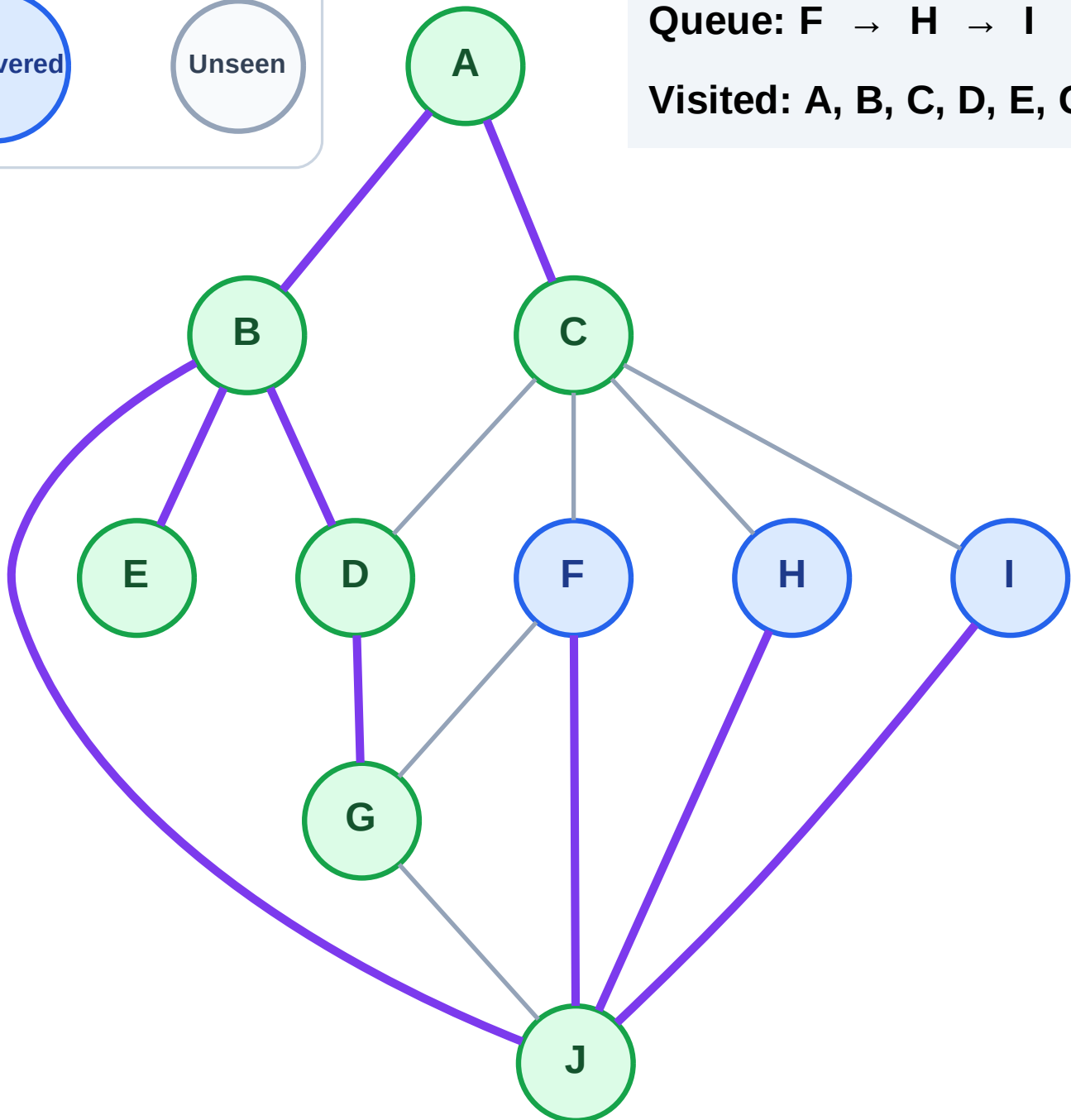
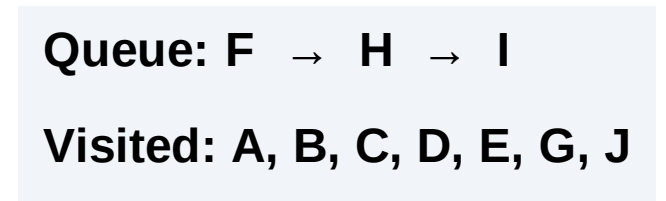


Queue: F → H → I
Visited: A, B, C, D, E, J

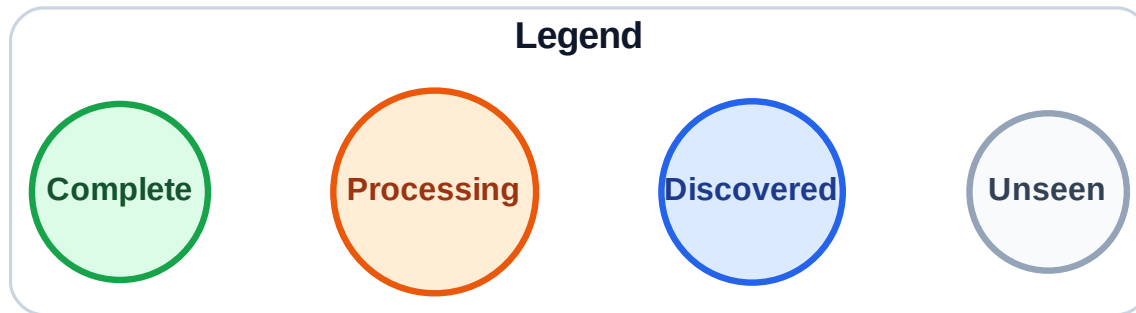


Step 15: No new neighbors from G

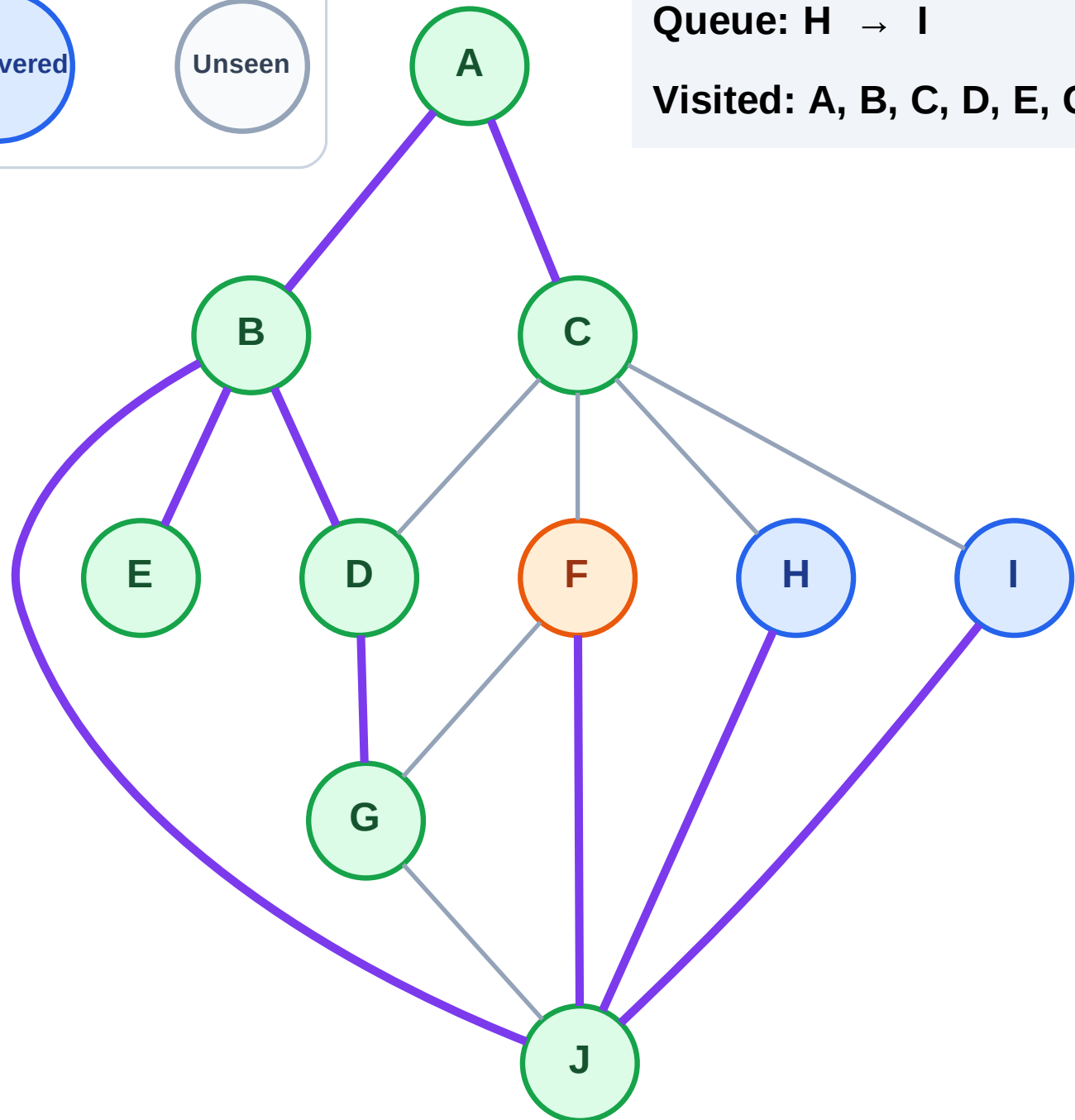
Step 15: No new neighbors from G



Step 16: Process node F



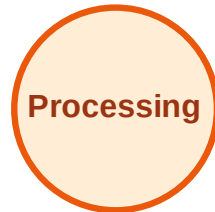
Visited: A, B, C, D, E, G, J



BFS Traversal

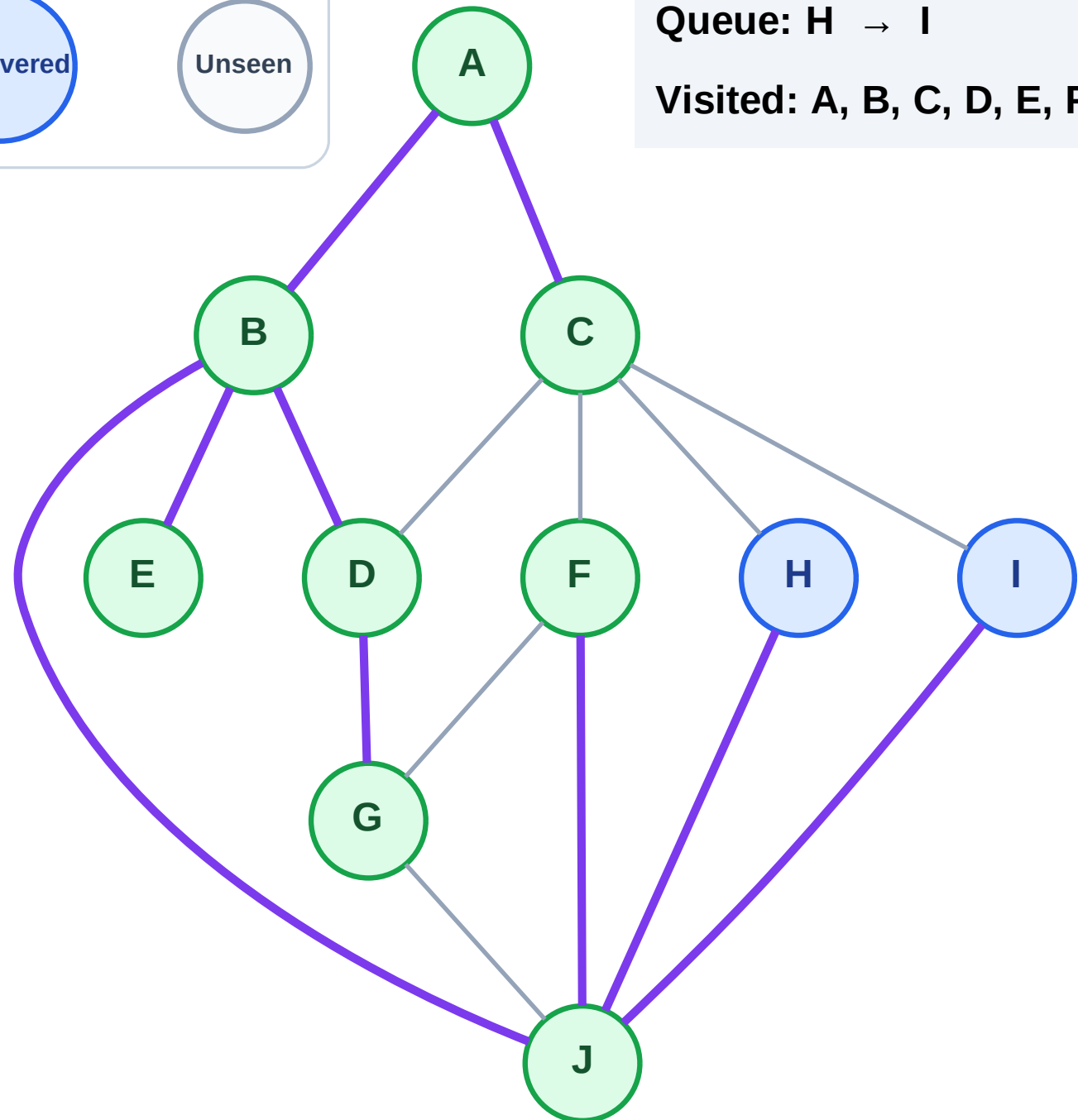
Step 17: No new neighbors from F

Legend



Queue: H → I

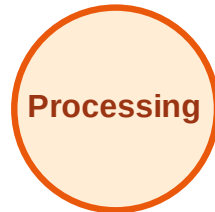
Visited: A, B, C, D, E, F, G, J



BFS Traversal

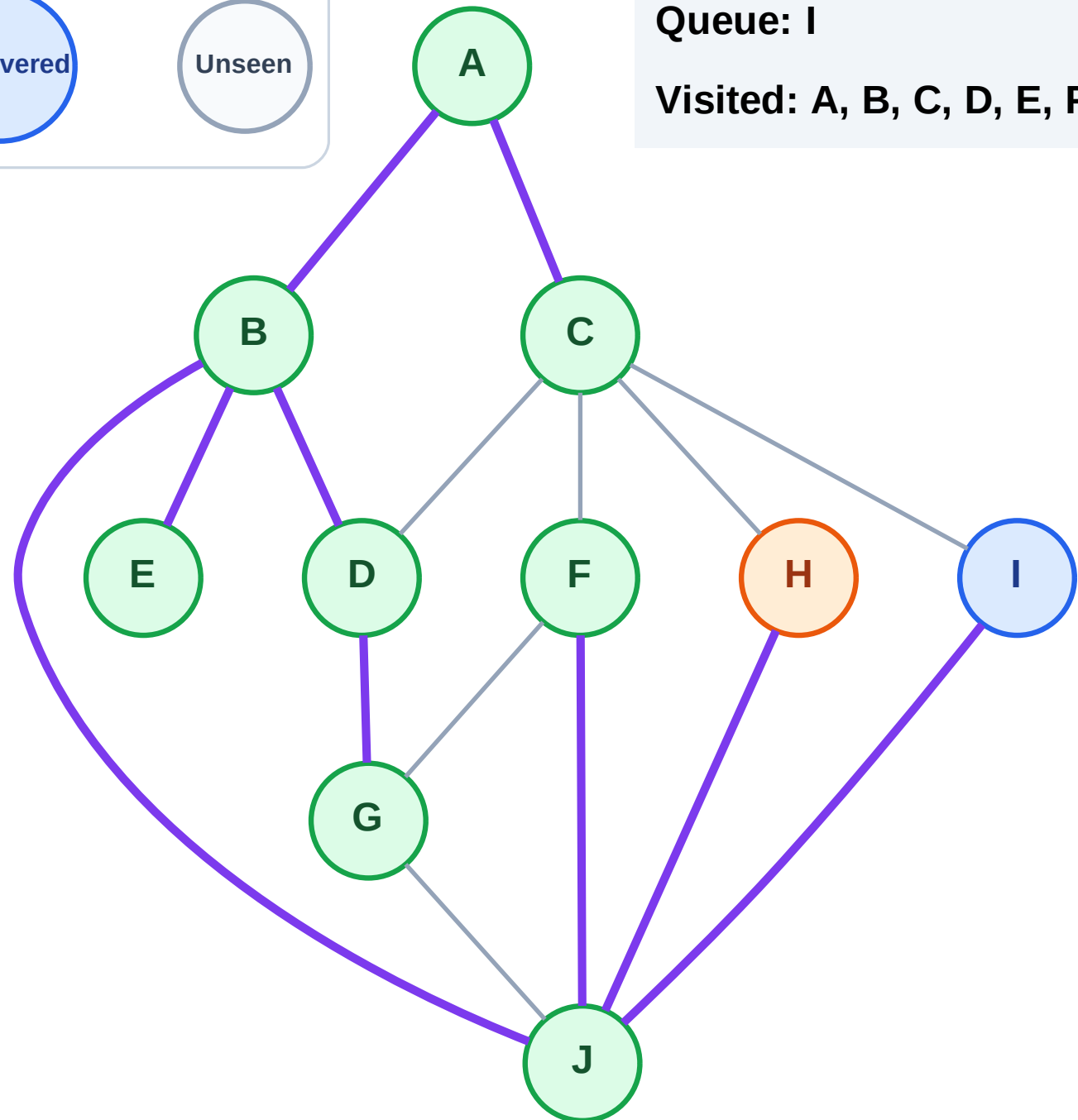
Step 18: Process node H

Legend



Queue: I

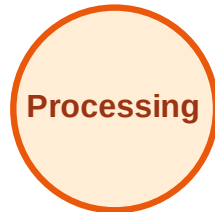
Visited: A, B, C, D, E, F, G, J



BFS Traversal

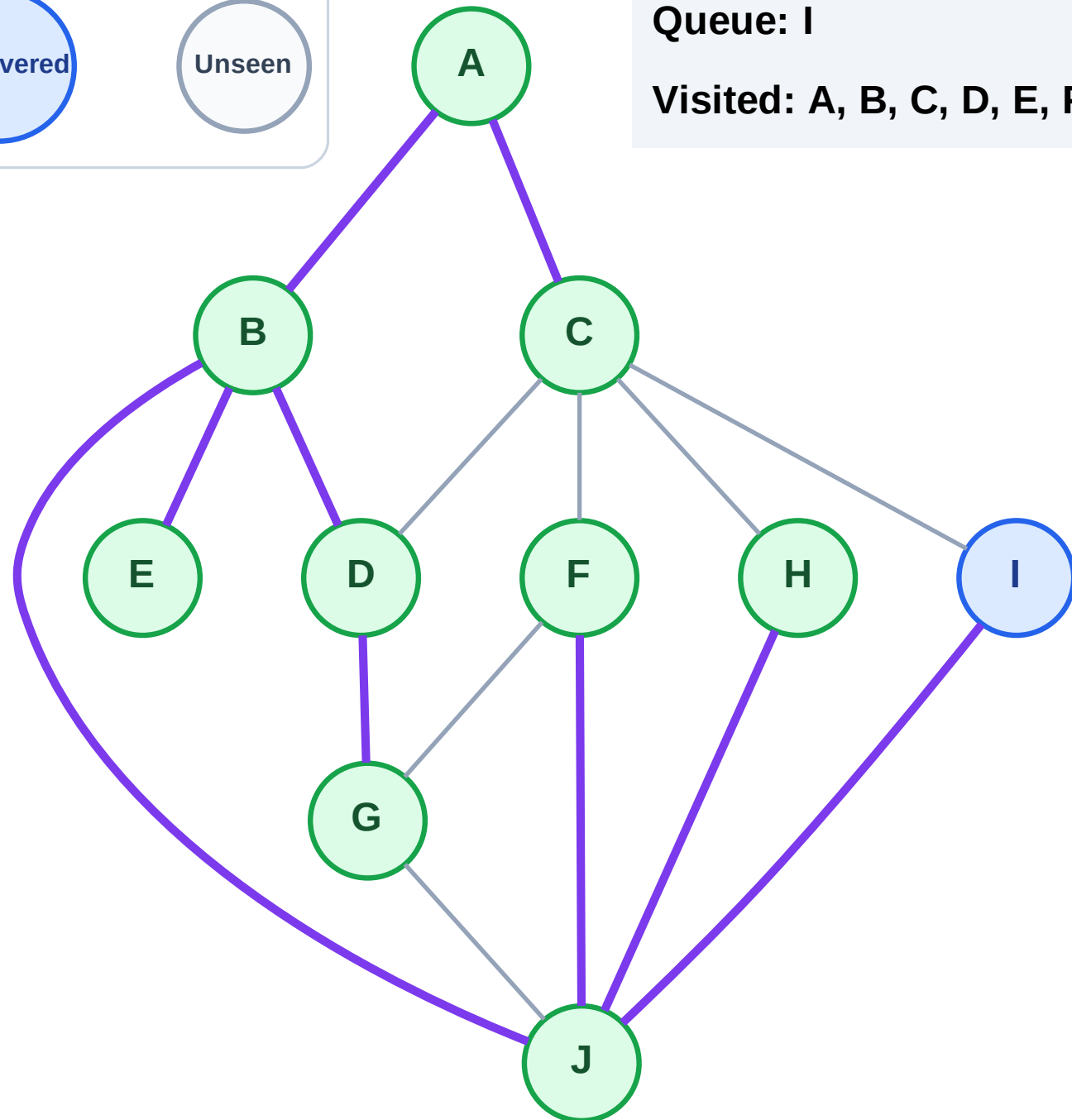
Step 19: No new neighbors from H

Legend



Queue: I

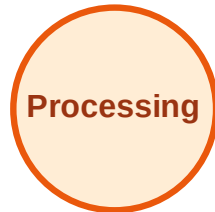
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

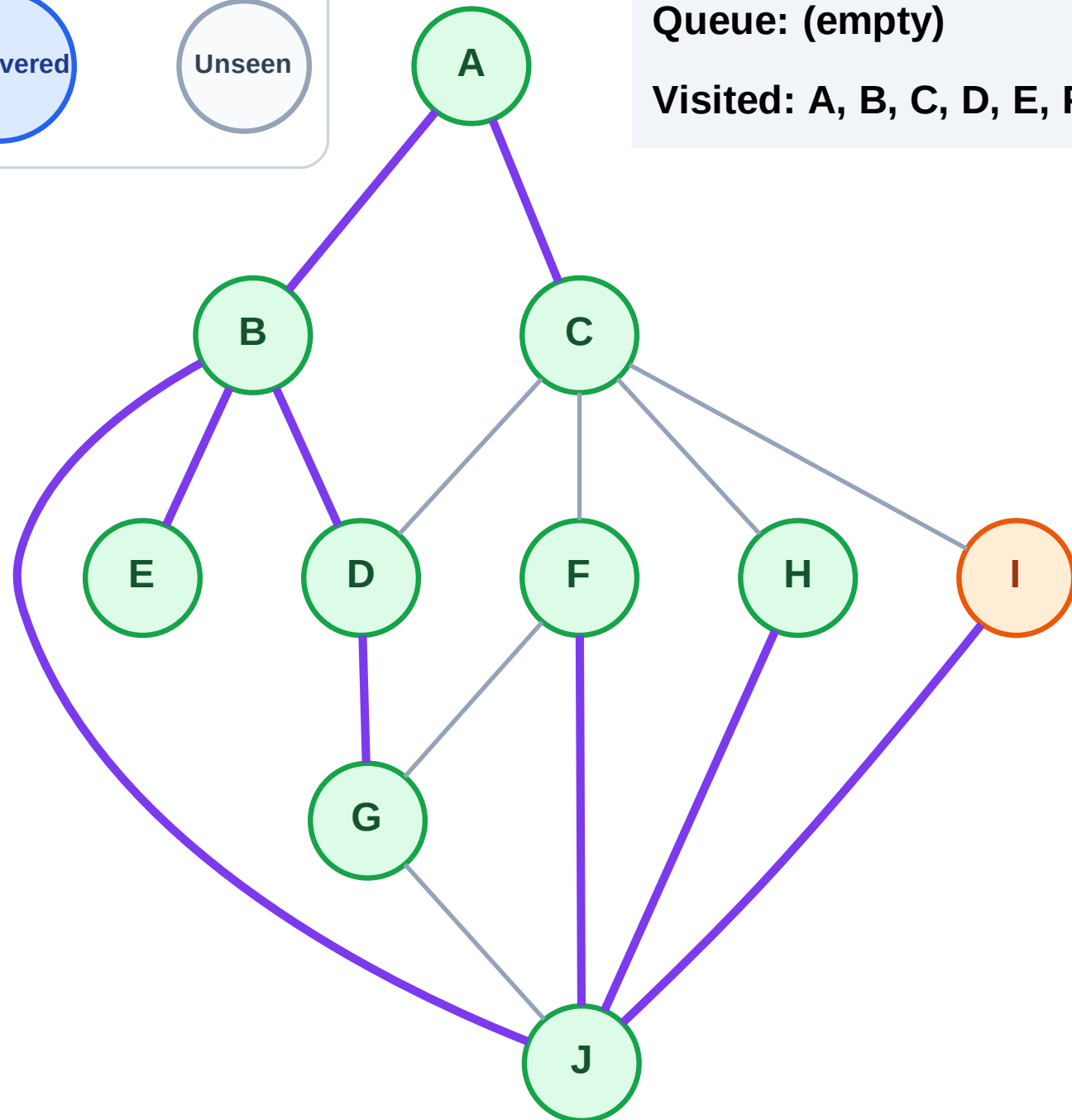
Step 20: Process node I

Legend



Queue: (empty)

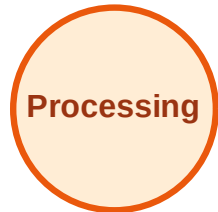
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

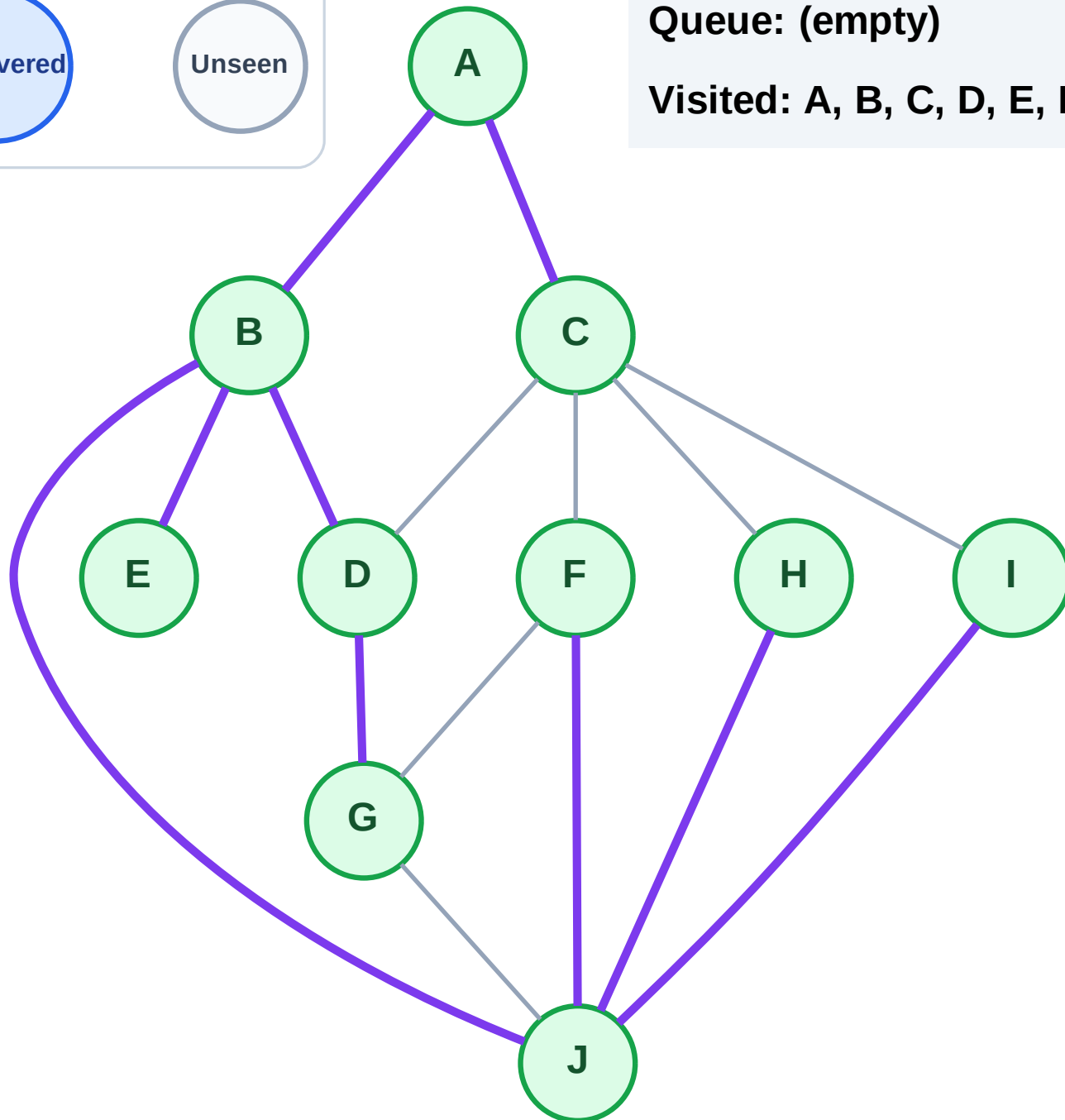
Step 21: No new neighbors from I

Legend



Queue: (empty)

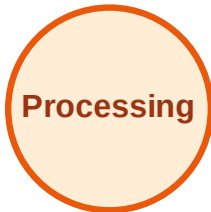
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

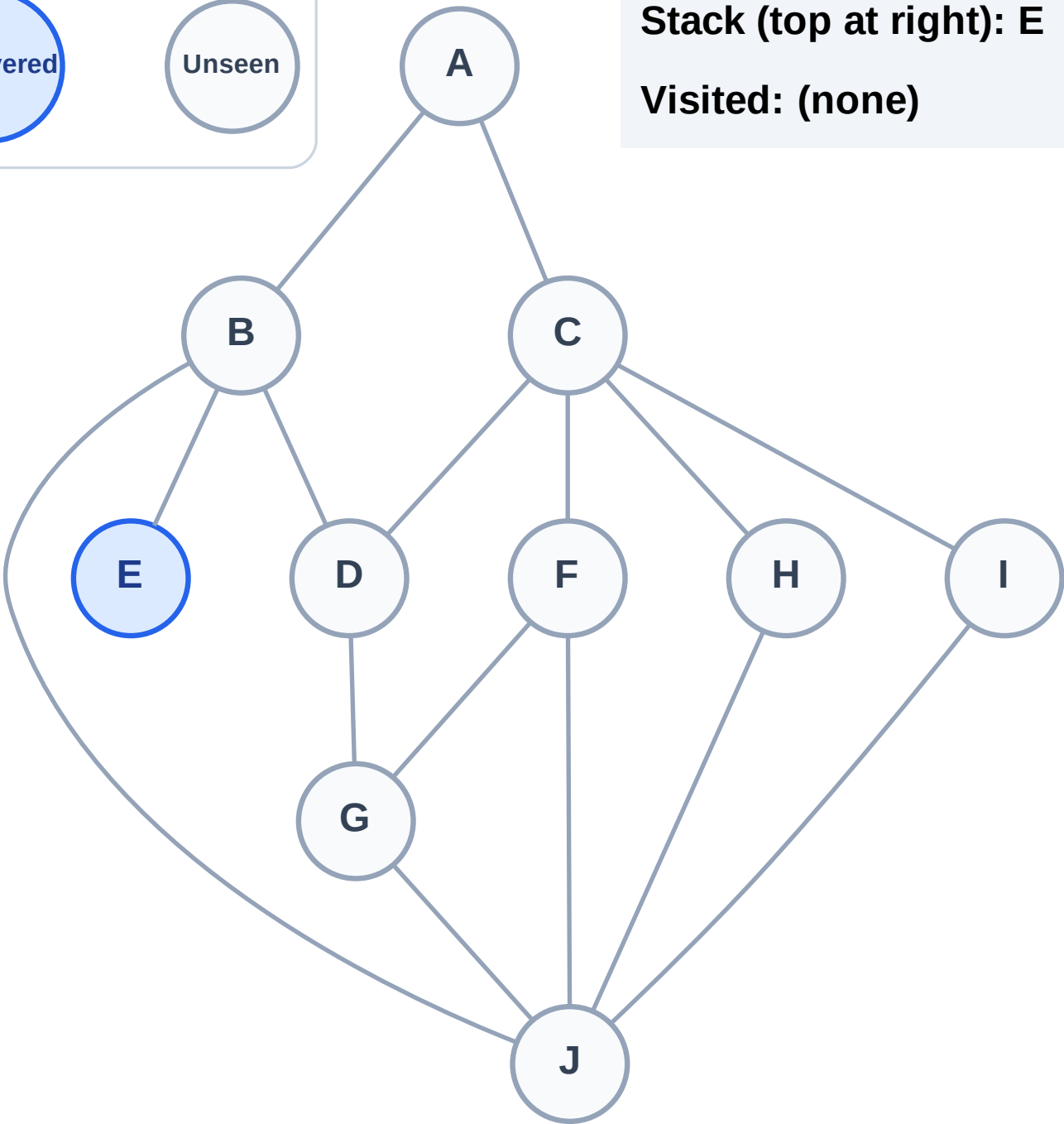
Step 1: Start at E

Legend



Stack (top at right): E

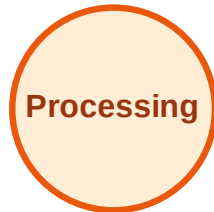
Visited: (none)



DFS Traversal

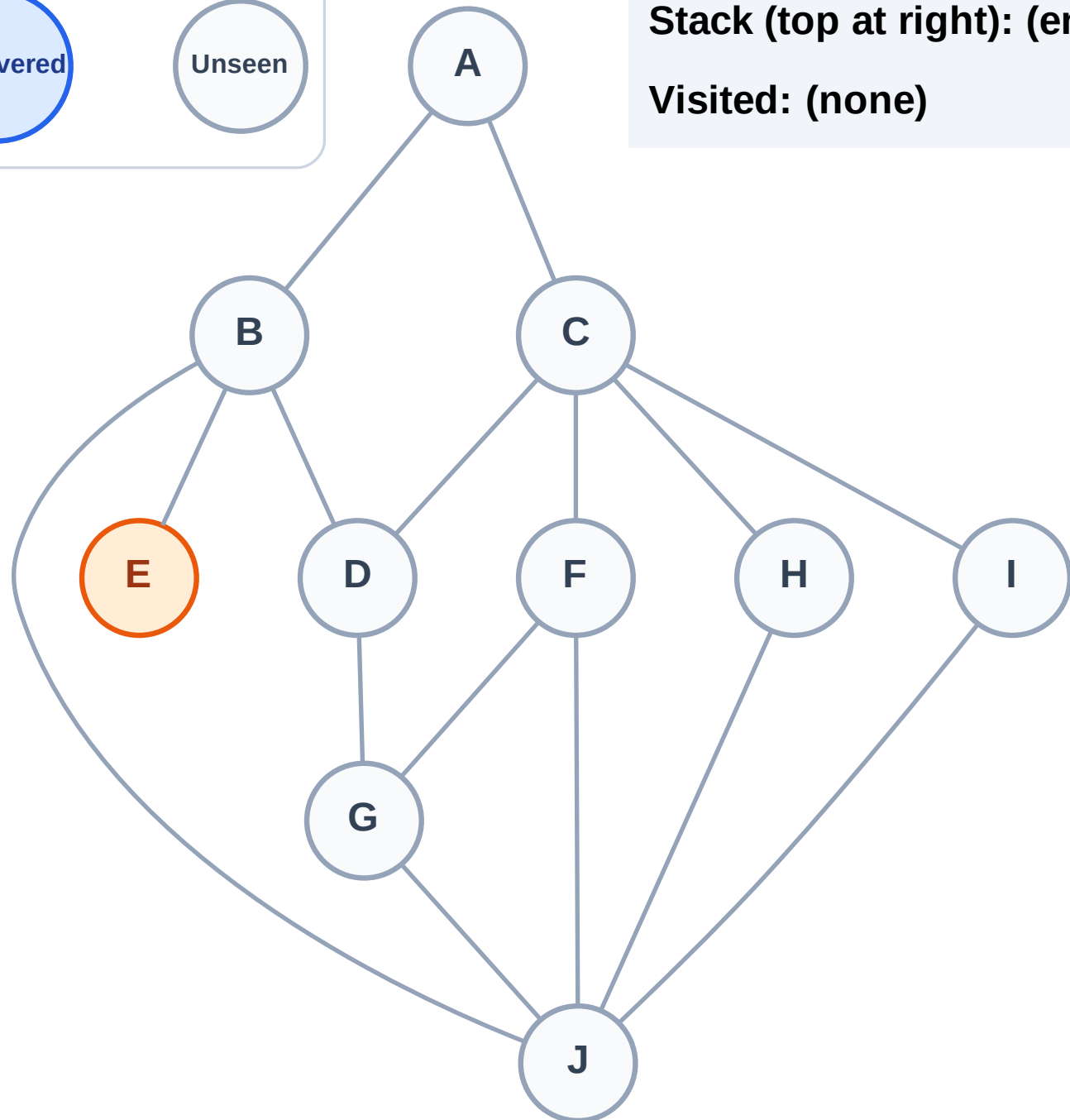
Step 2: Process node E

Legend



Stack (top at right): (empty)

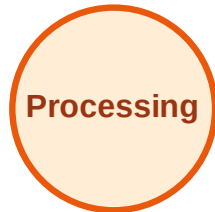
Visited: (none)



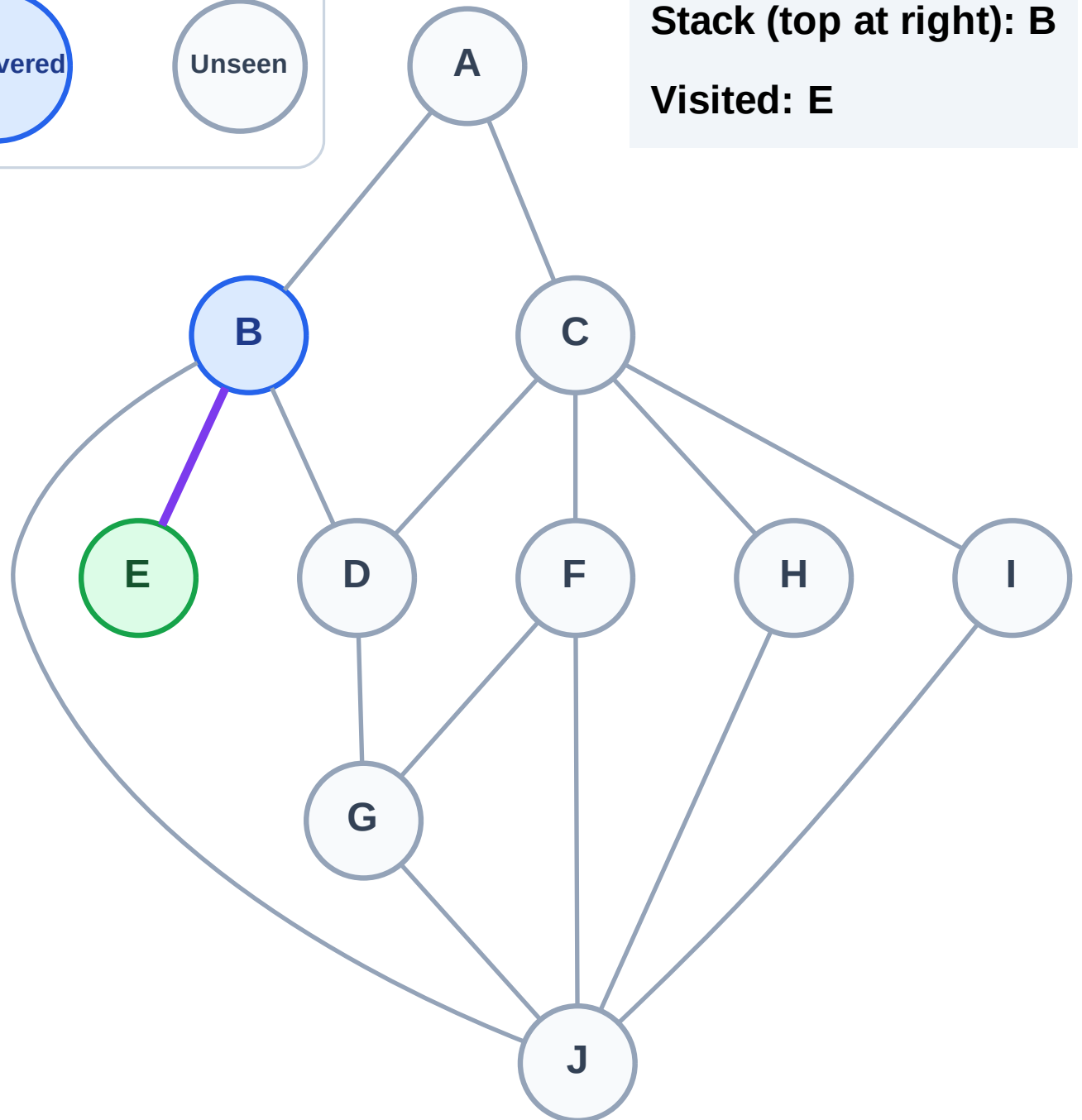
DFS Traversal

Step 3: Discover B from E

Legend



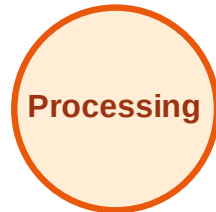
Stack (top at right): B
Visited: E



DFS Traversal

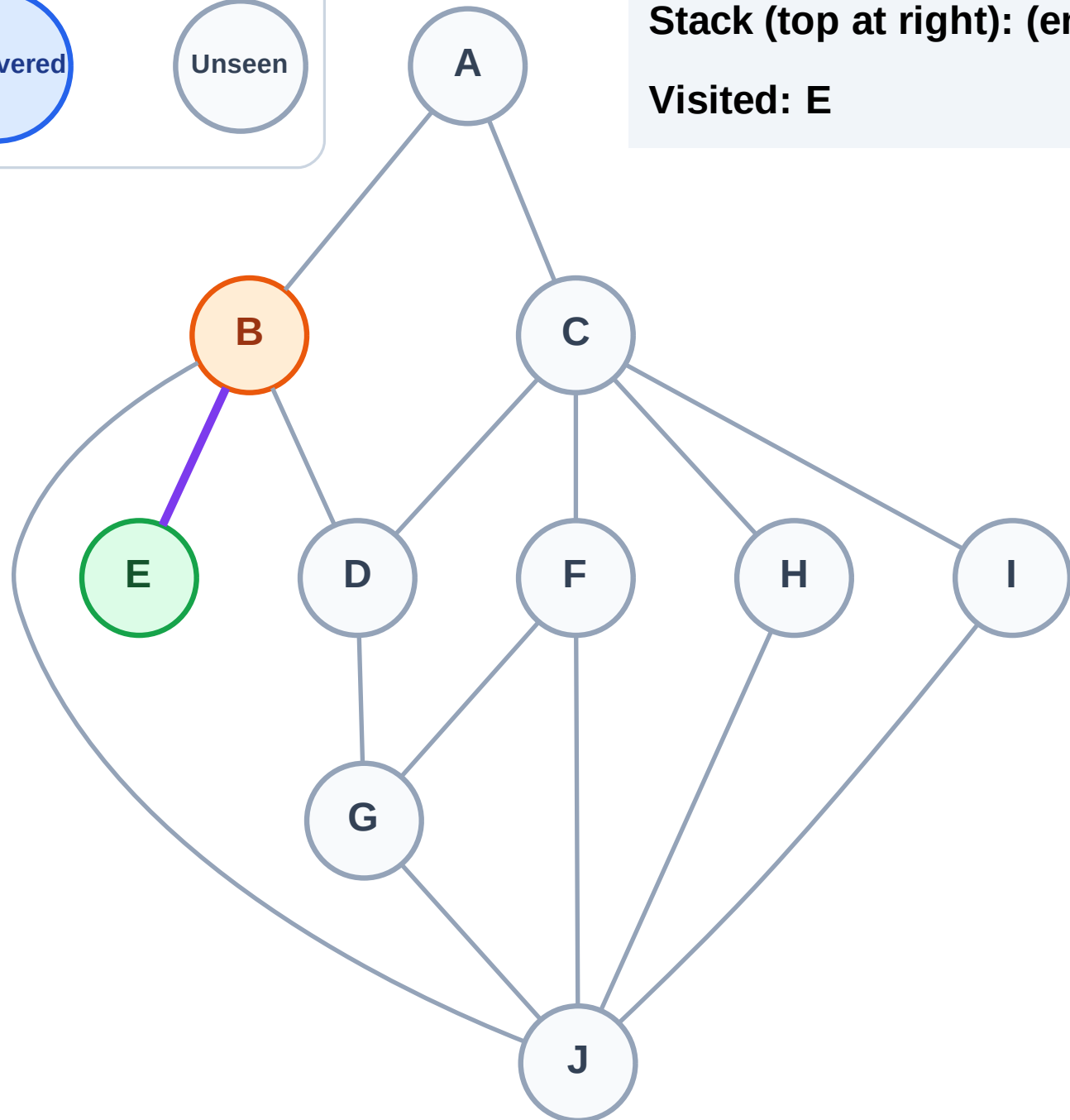
Step 4: Process node B

Legend



Stack (top at right): (empty)

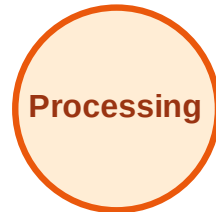
Visited: E



DFS Traversal

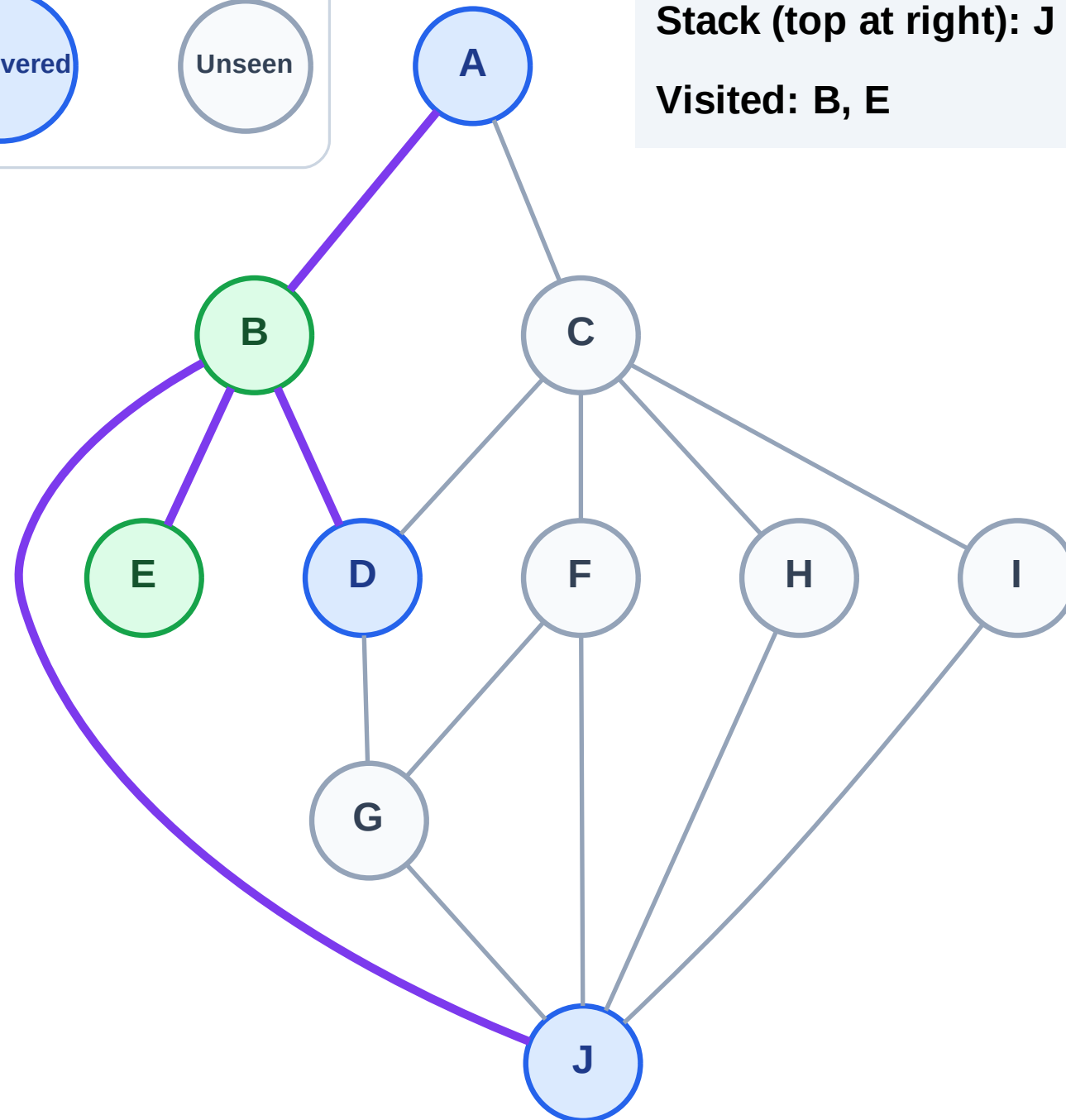
Step 5: Discover J, D, A from B

Legend



Stack (top at right): J | D | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

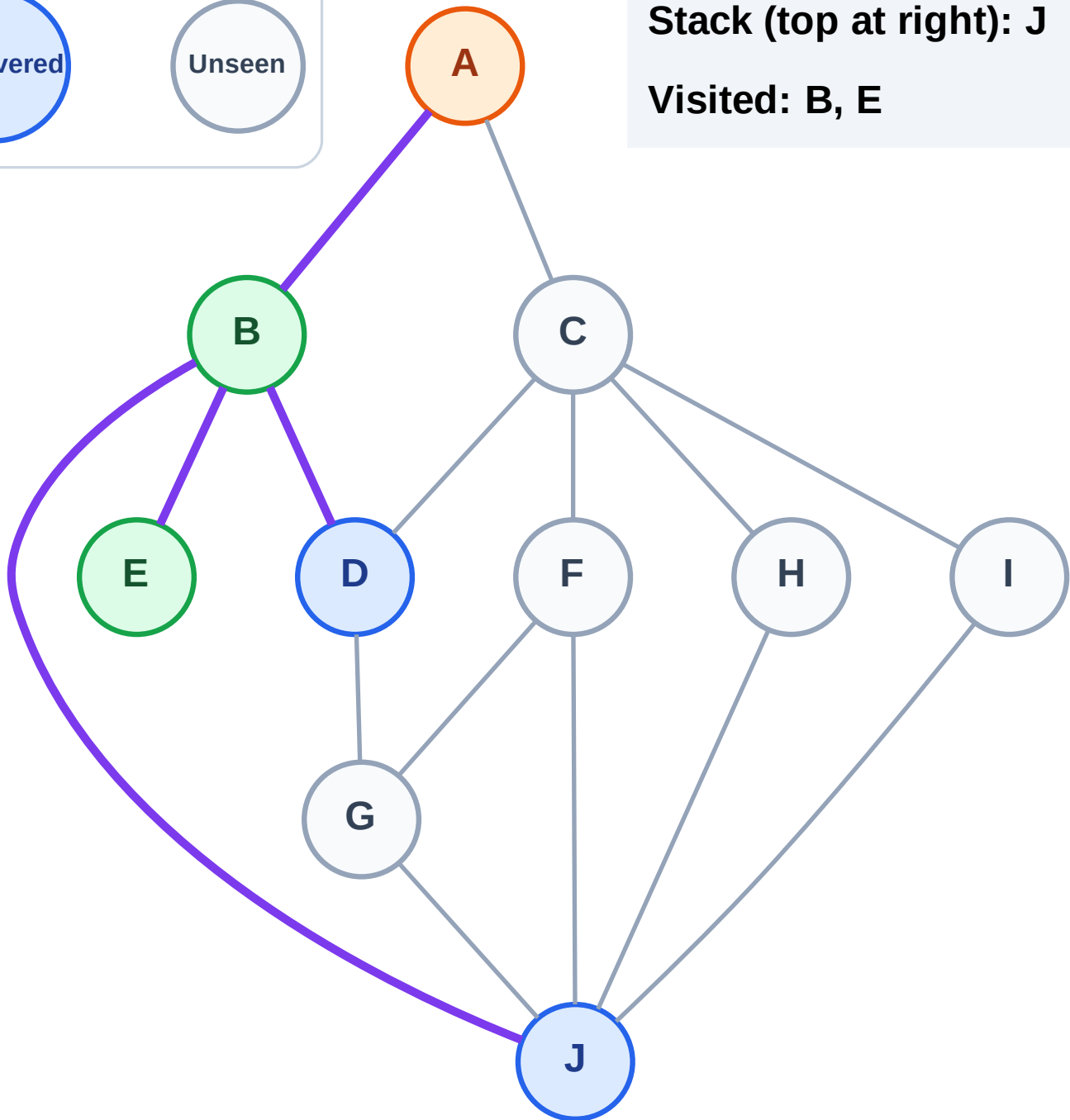
Processing

Discovered

Unseen

Stack (top at right): J | D

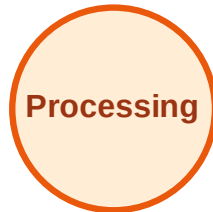
Visited: B, E



DFS Traversal

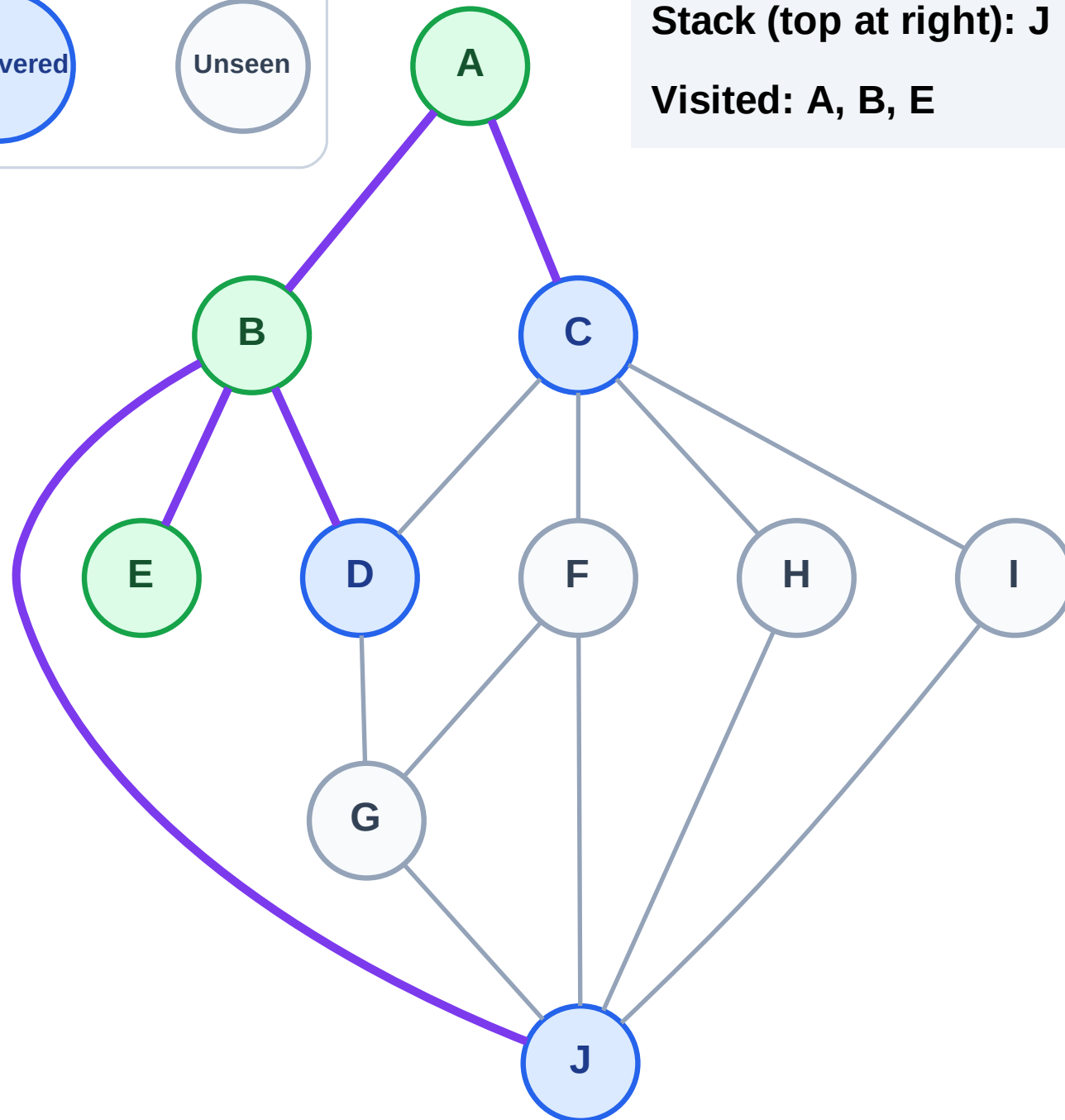
Step 7: Discover C from A

Legend



Stack (top at right): J | D | C

Visited: A, B, E



DFS Traversal

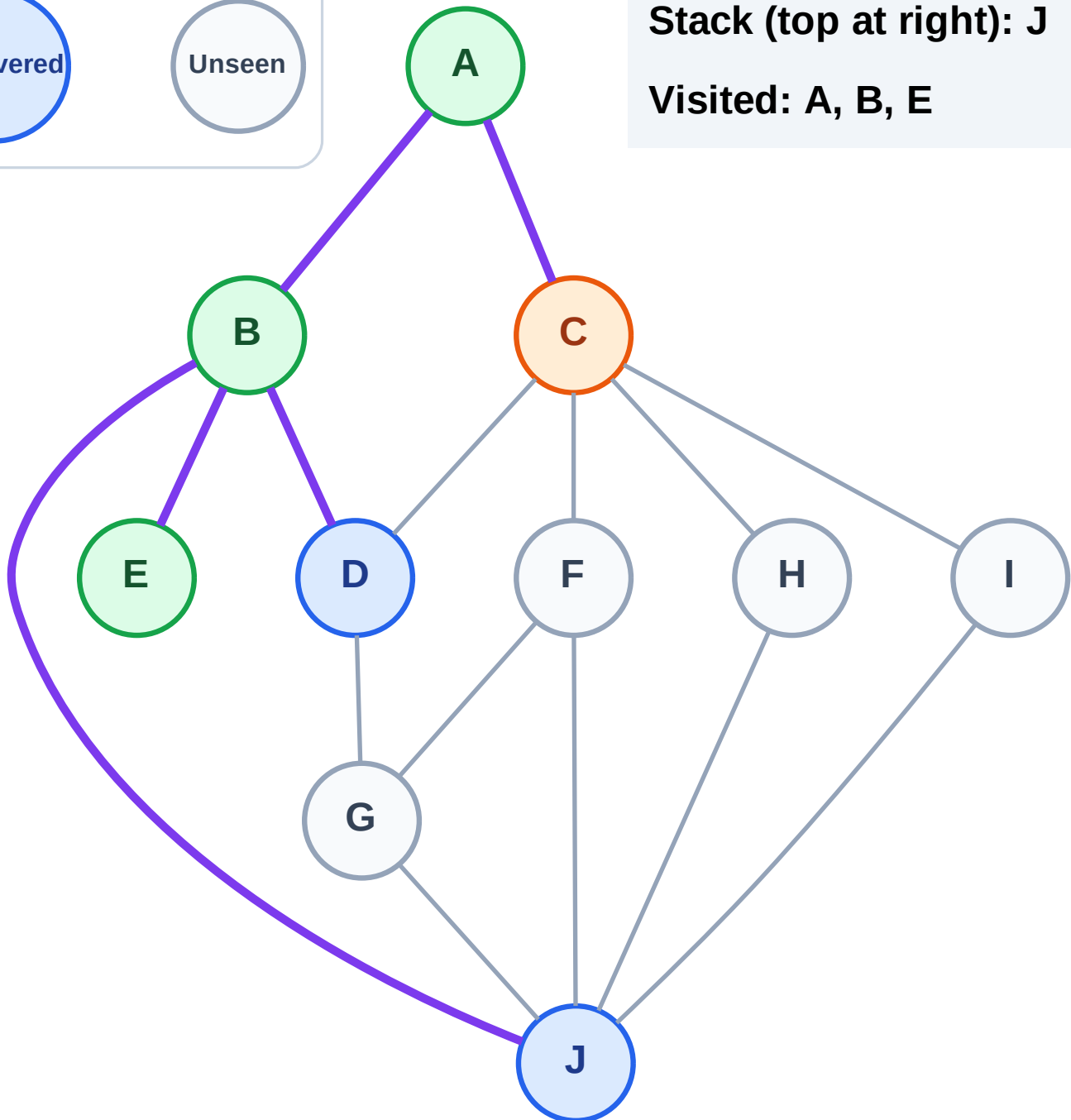
Step 8: Process node C

Legend



Stack (top at right): J | D

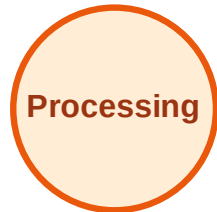
Visited: A, B, E



DFS Traversal

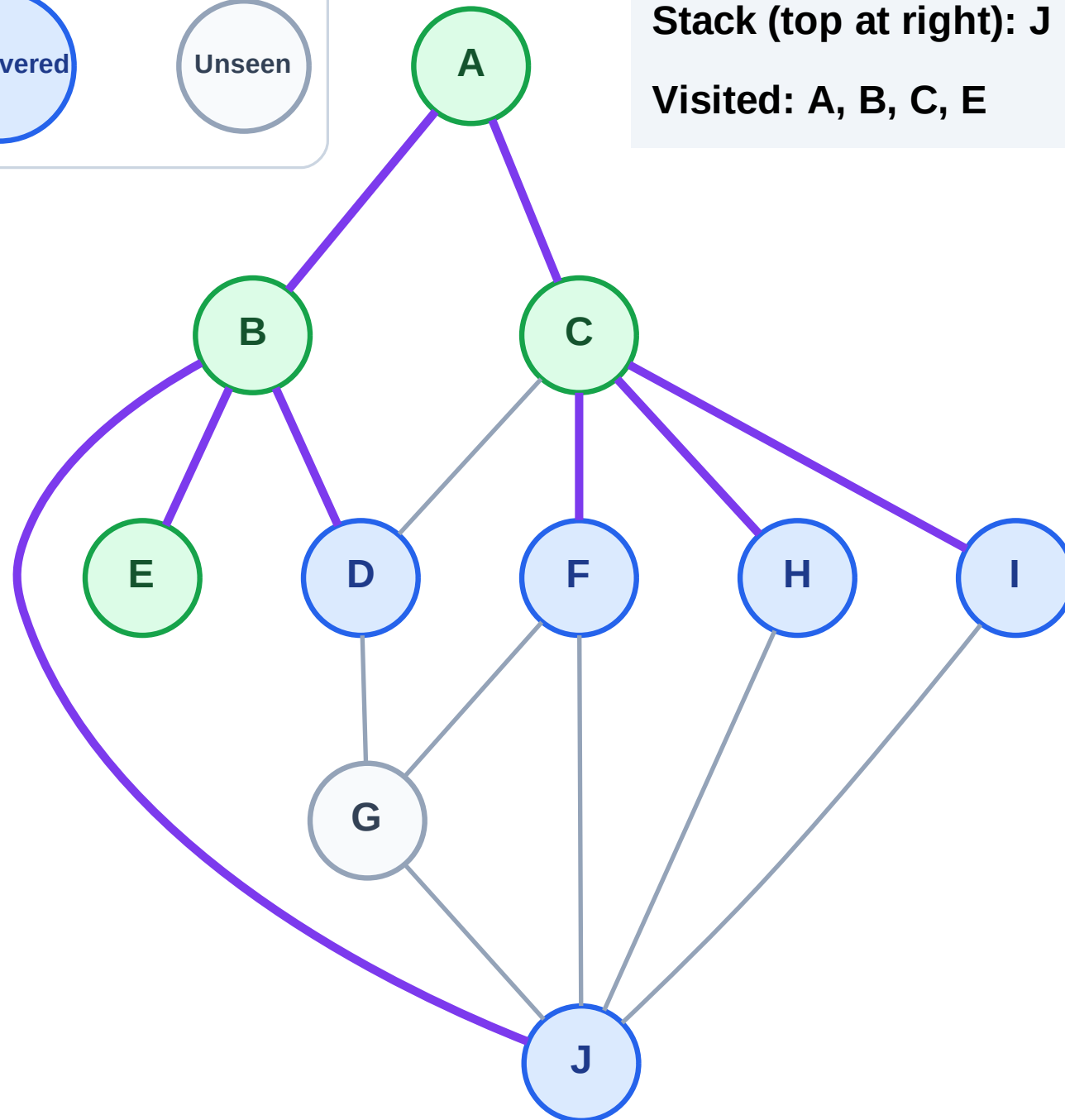
Step 9: Discover I, H, F from C

Legend



Stack (top at right): J | D | I | H | F

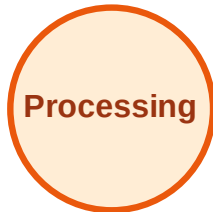
Visited: A, B, C, E



DFS Traversal

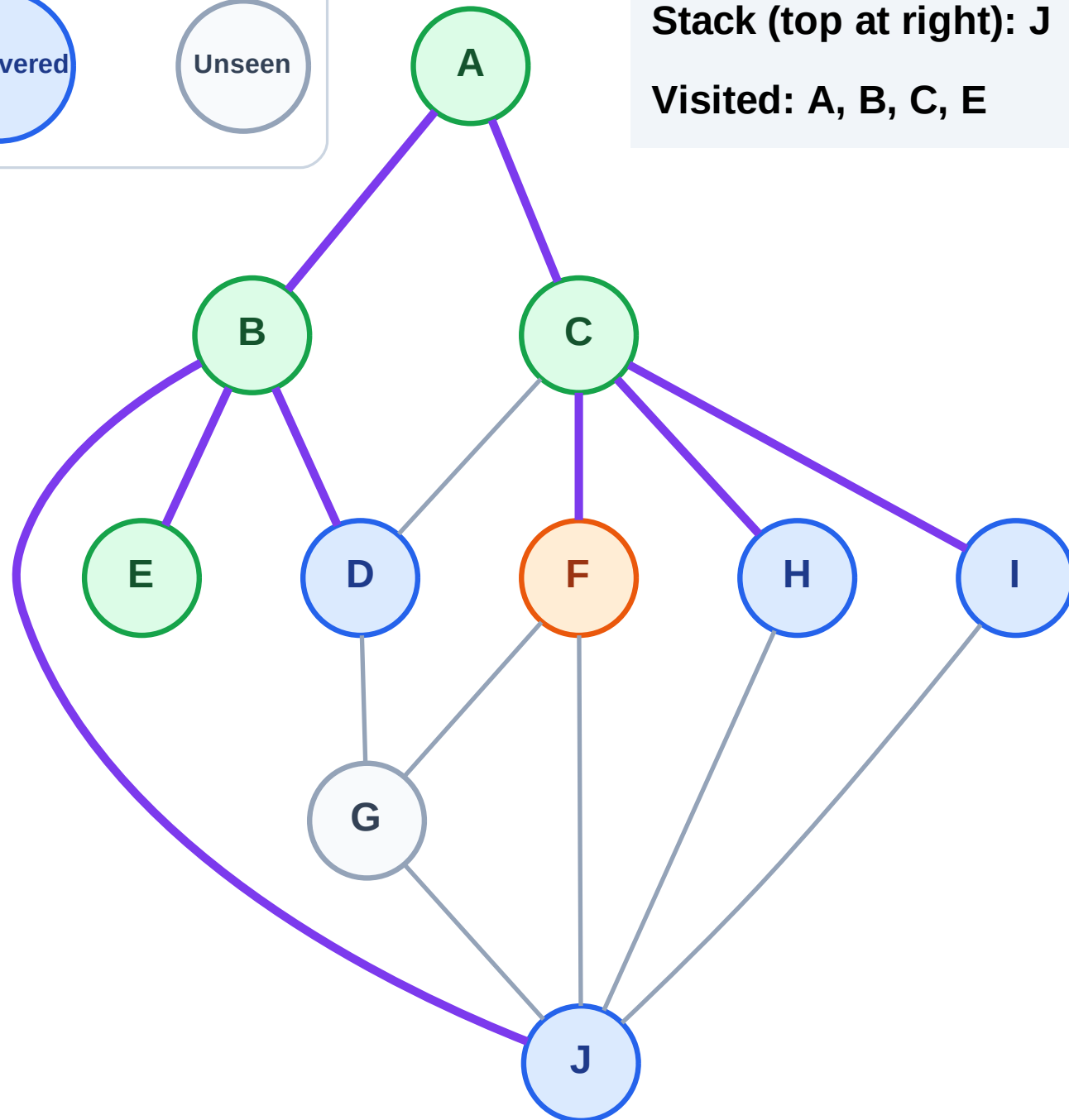
Step 10: Process node F

Legend



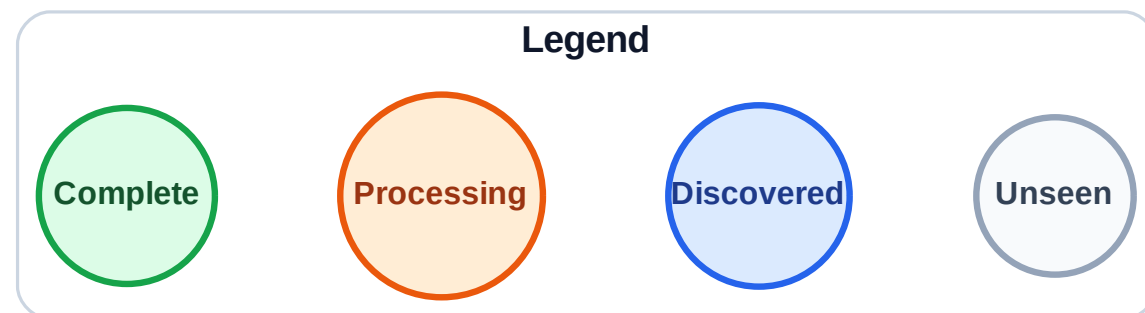
Stack (top at right): J | D | I | H

Visited: A, B, C, E

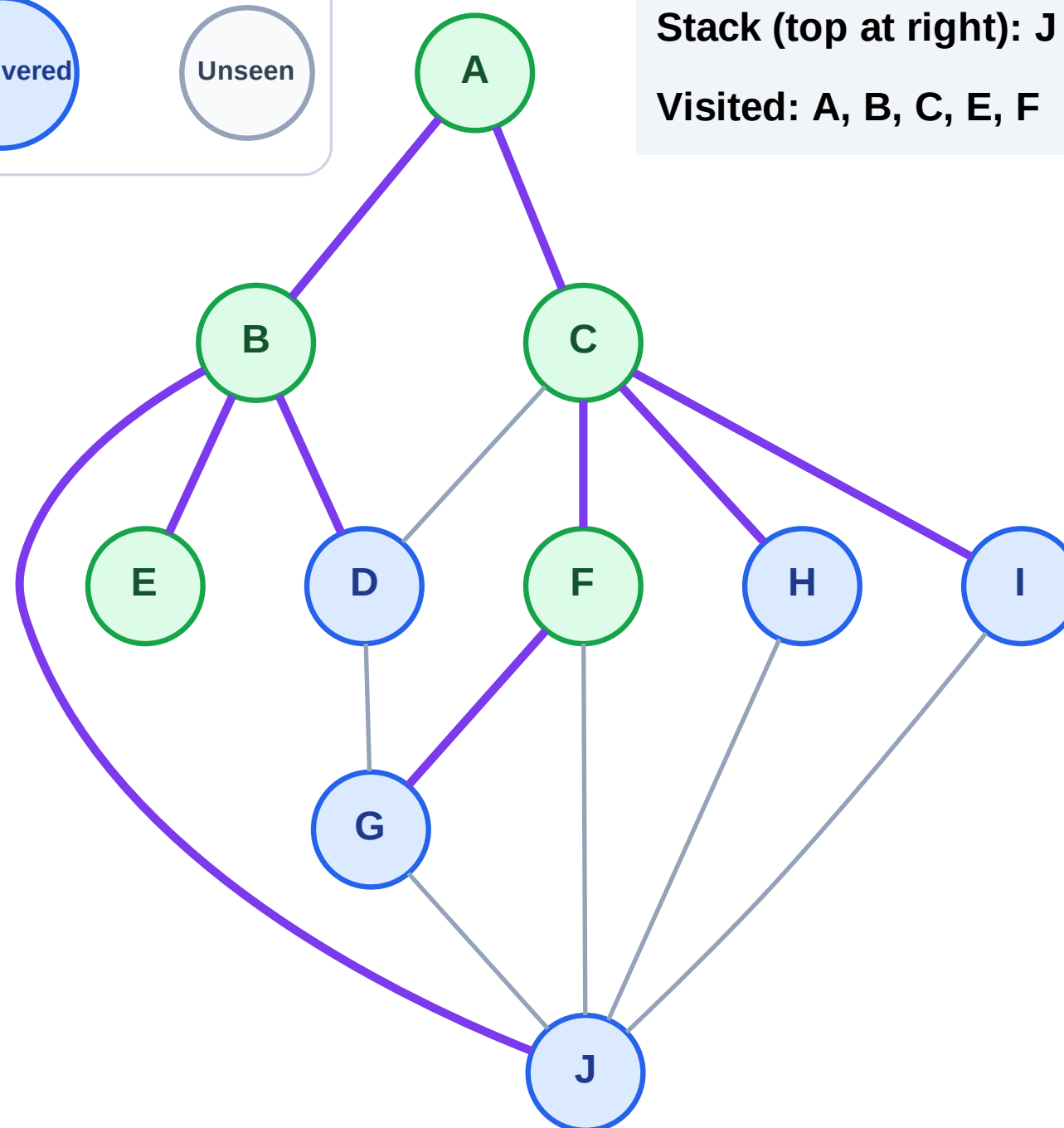


DFS Traversal

Step 11: Discover G from F



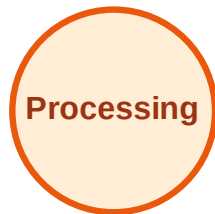
Stack (top at right): J | D | I | H | G
Visited: A, B, C, E, F



DFS Traversal

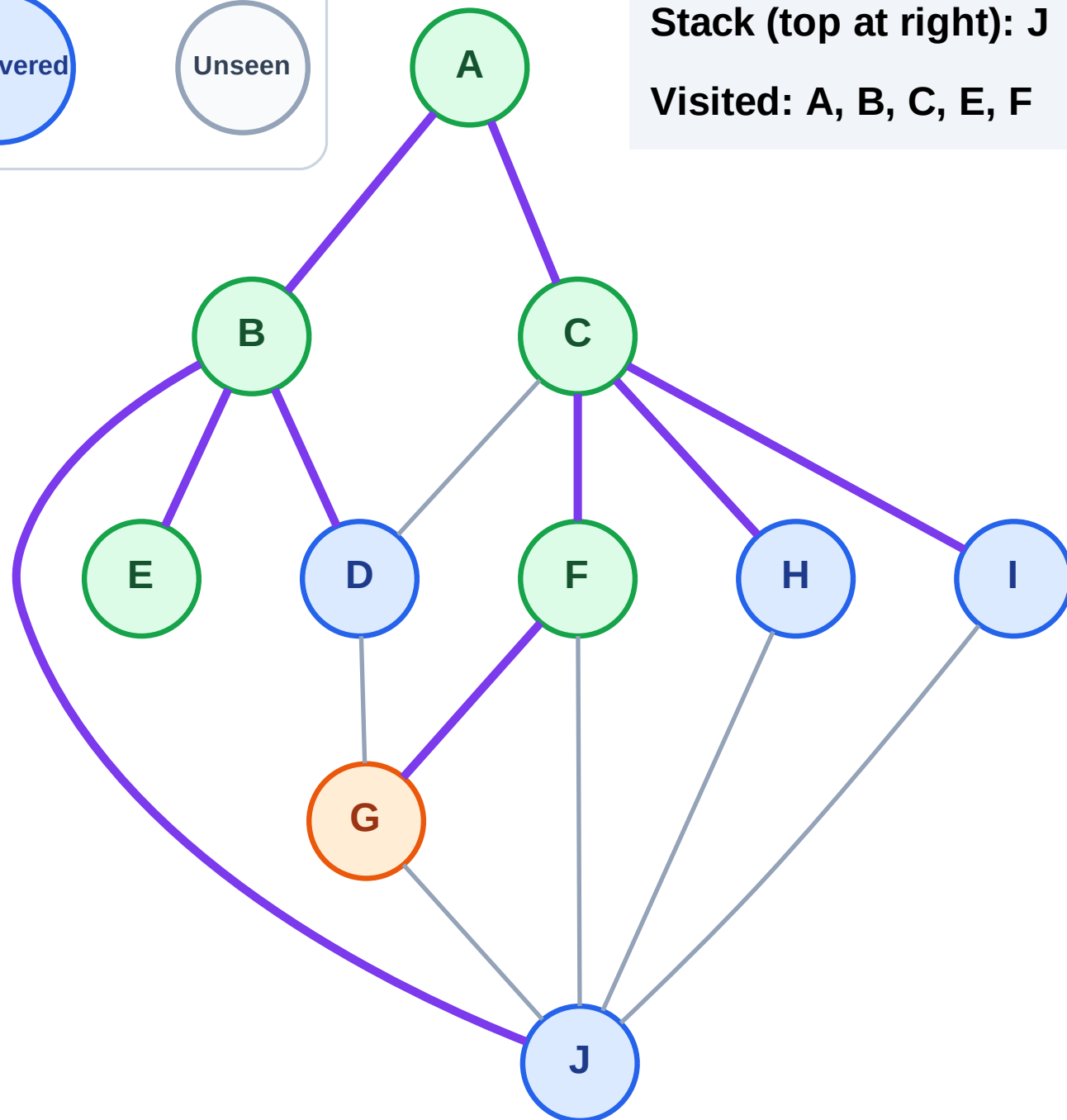
Step 12: Process node G

Legend



Stack (top at right): J | D | I | H

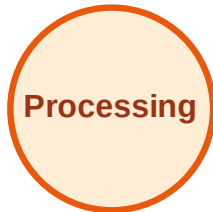
Visited: A, B, C, E, F



DFS Traversal

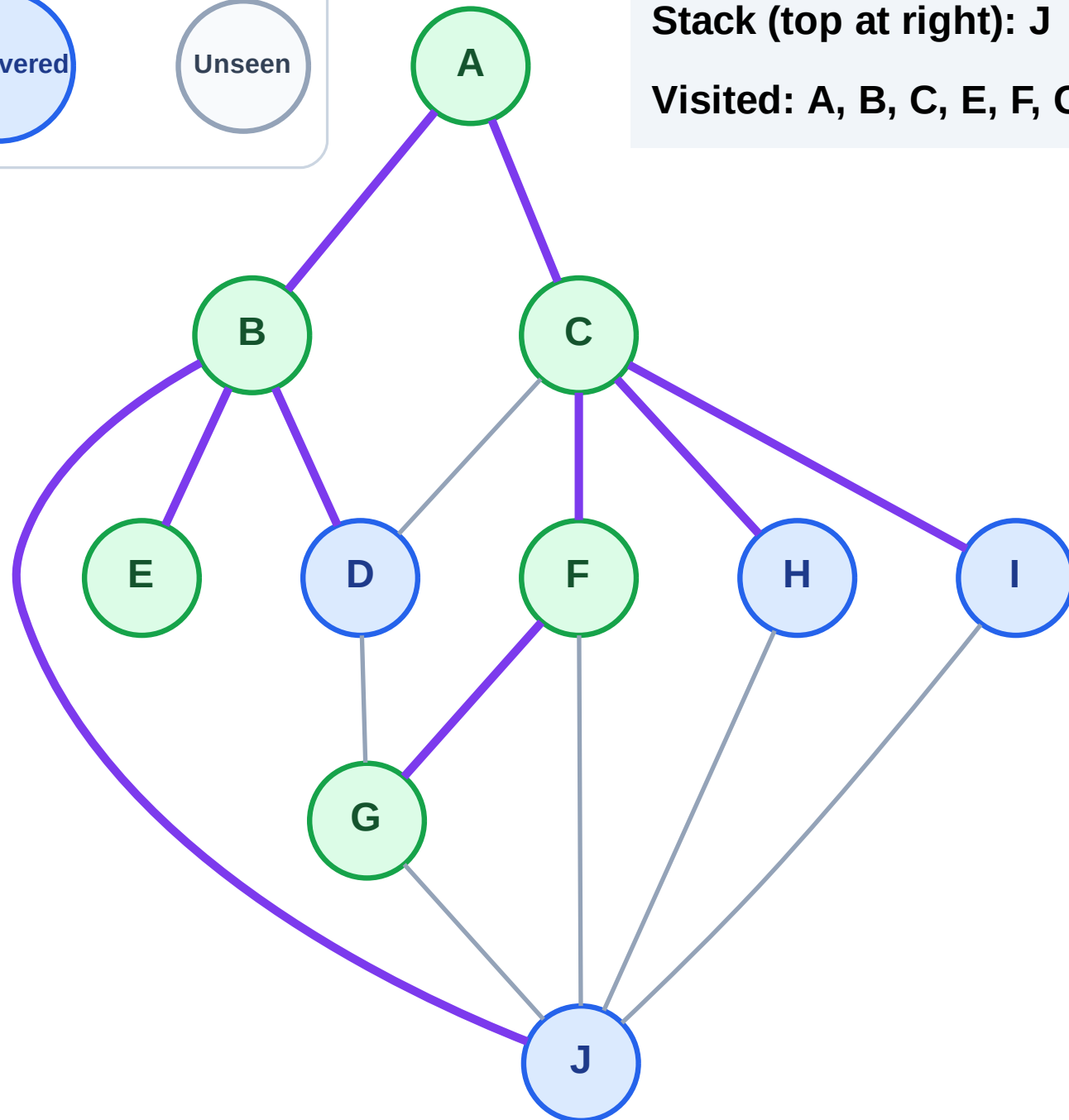
Step 13: No new neighbors from G

Legend



Stack (top at right): J | D | I | H

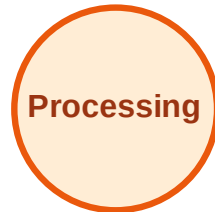
Visited: A, B, C, E, F, G



DFS Traversal

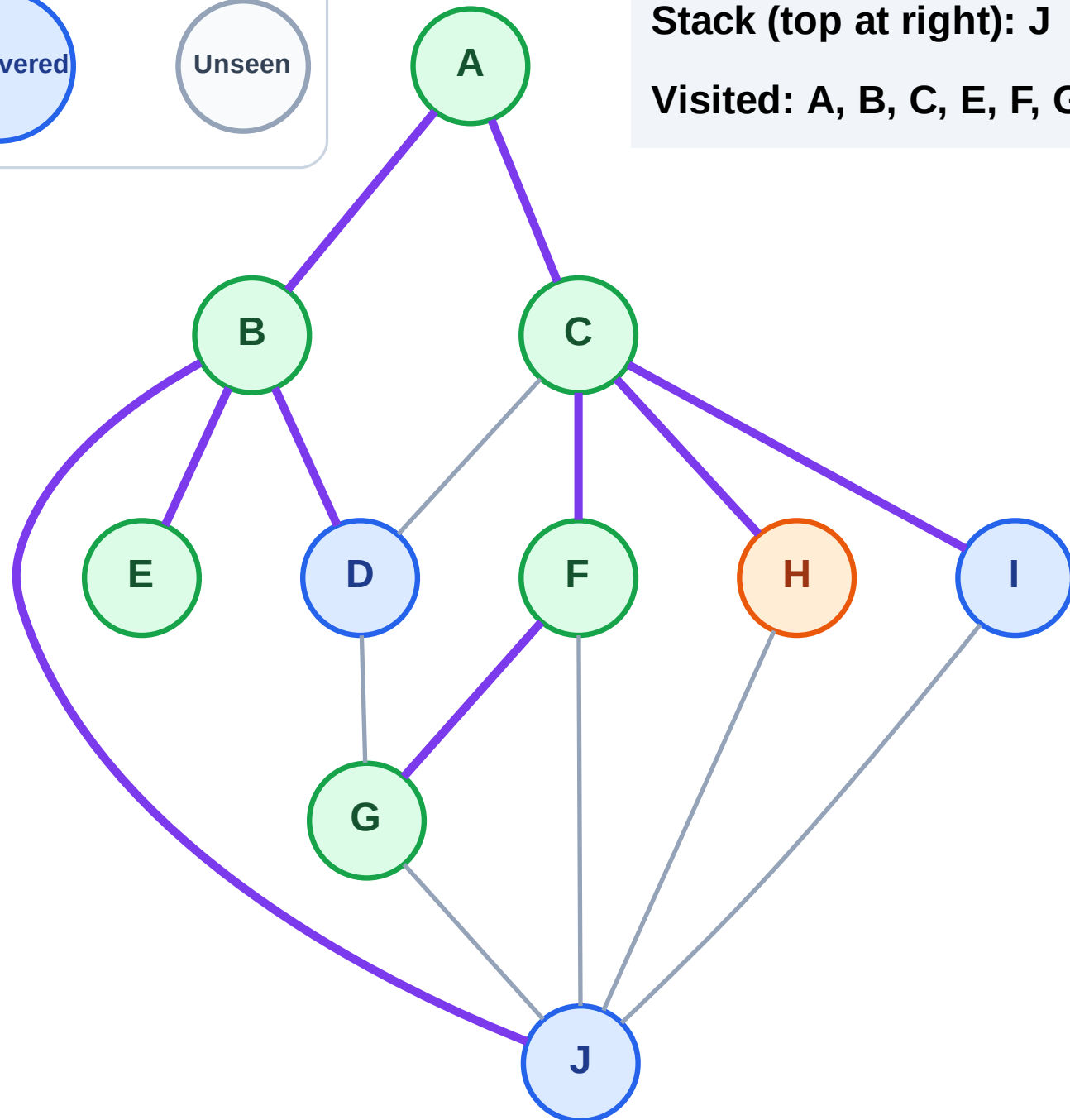
Step 14: Process node H

Legend



Stack (top at right): J | D | I

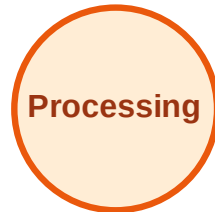
Visited: A, B, C, E, F, G



DFS Traversal

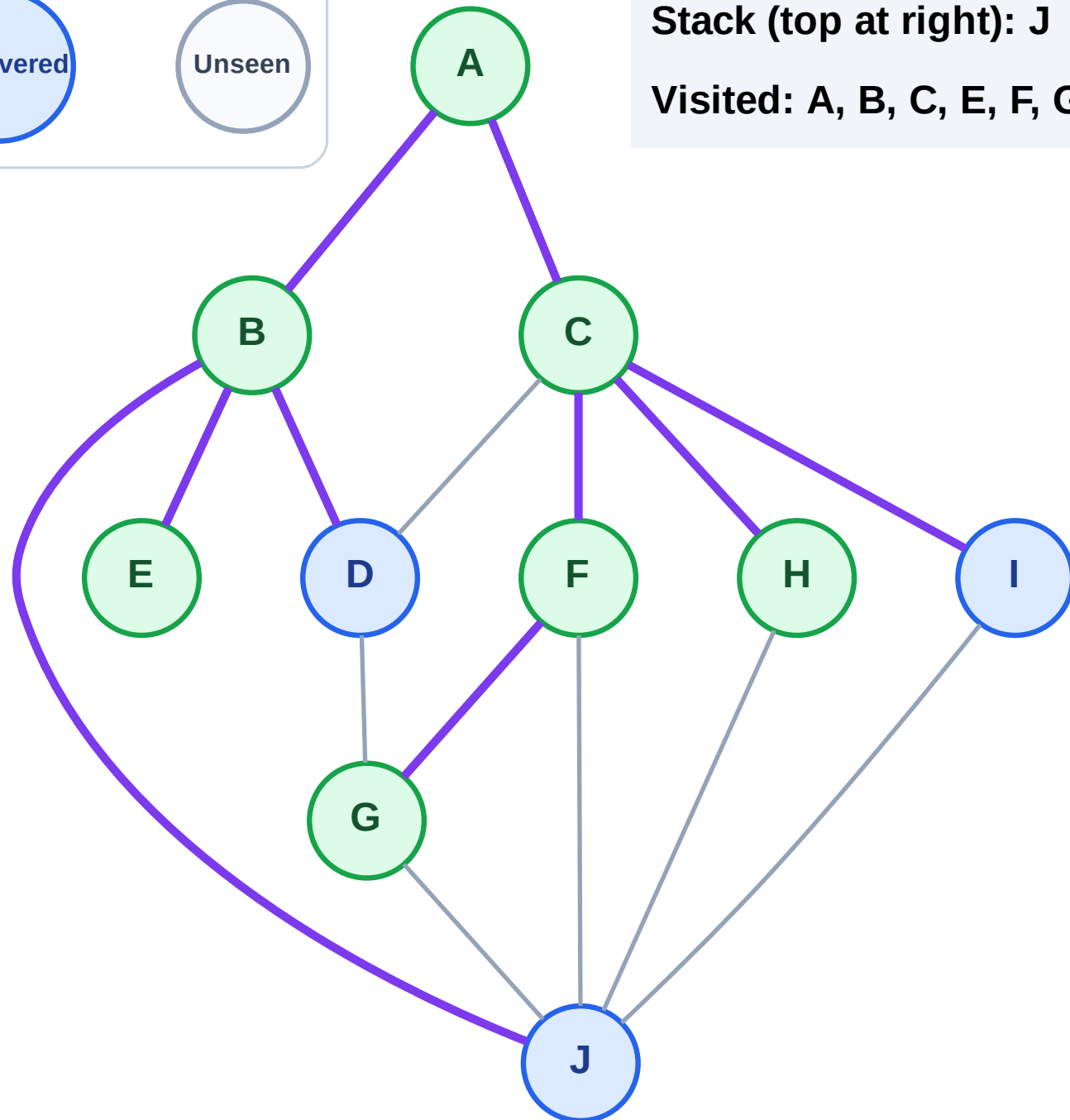
Step 15: No new neighbors from H

Legend



Stack (top at right): J | D | I

Visited: A, B, C, E, F, G, H



DFS Traversal

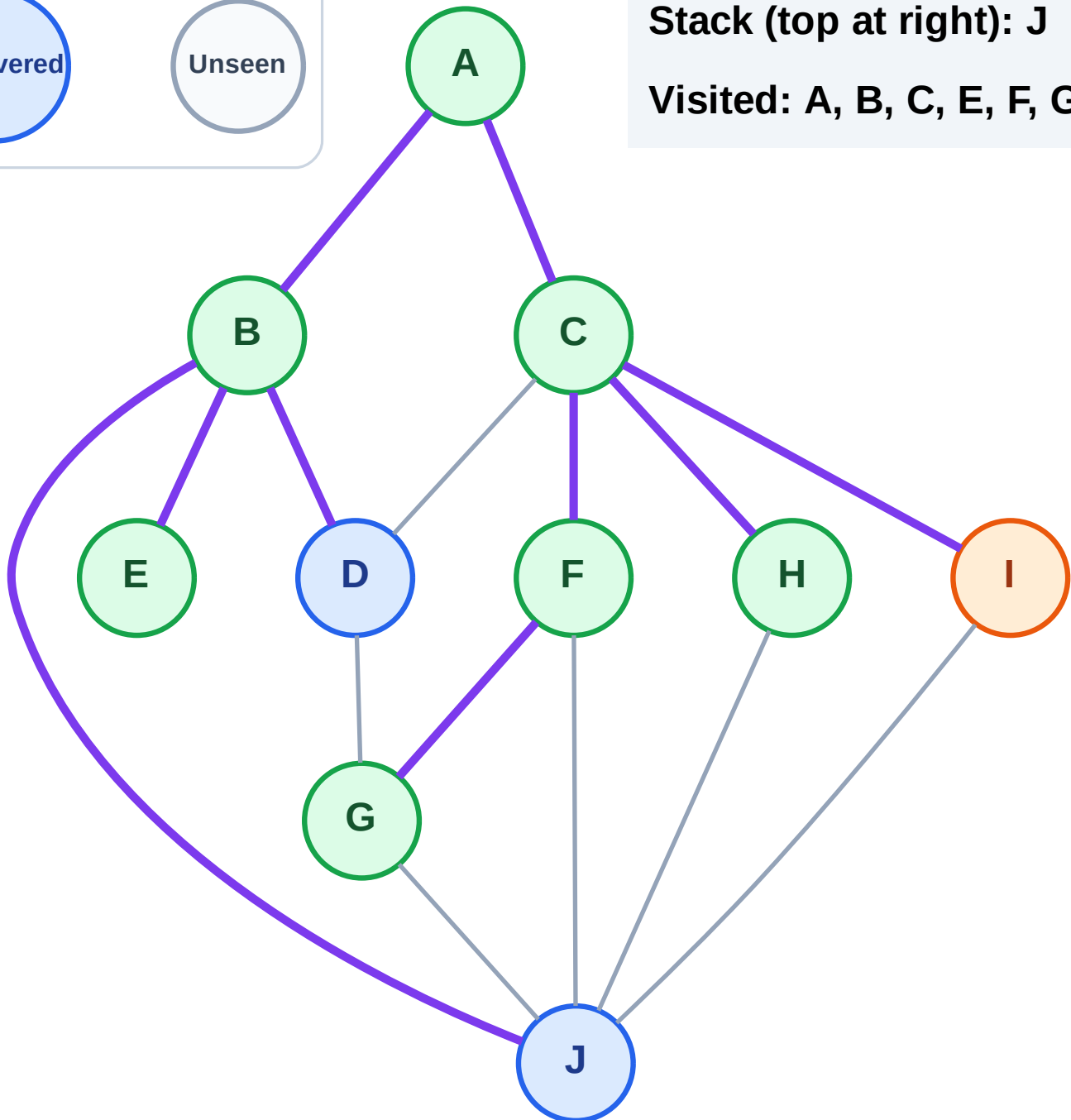
Step 16: Process node I

Legend



Stack (top at right): J | D

Visited: A, B, C, E, F, G, H



DFS Traversal

Step 17: No new neighbors from I

Legend

Complete

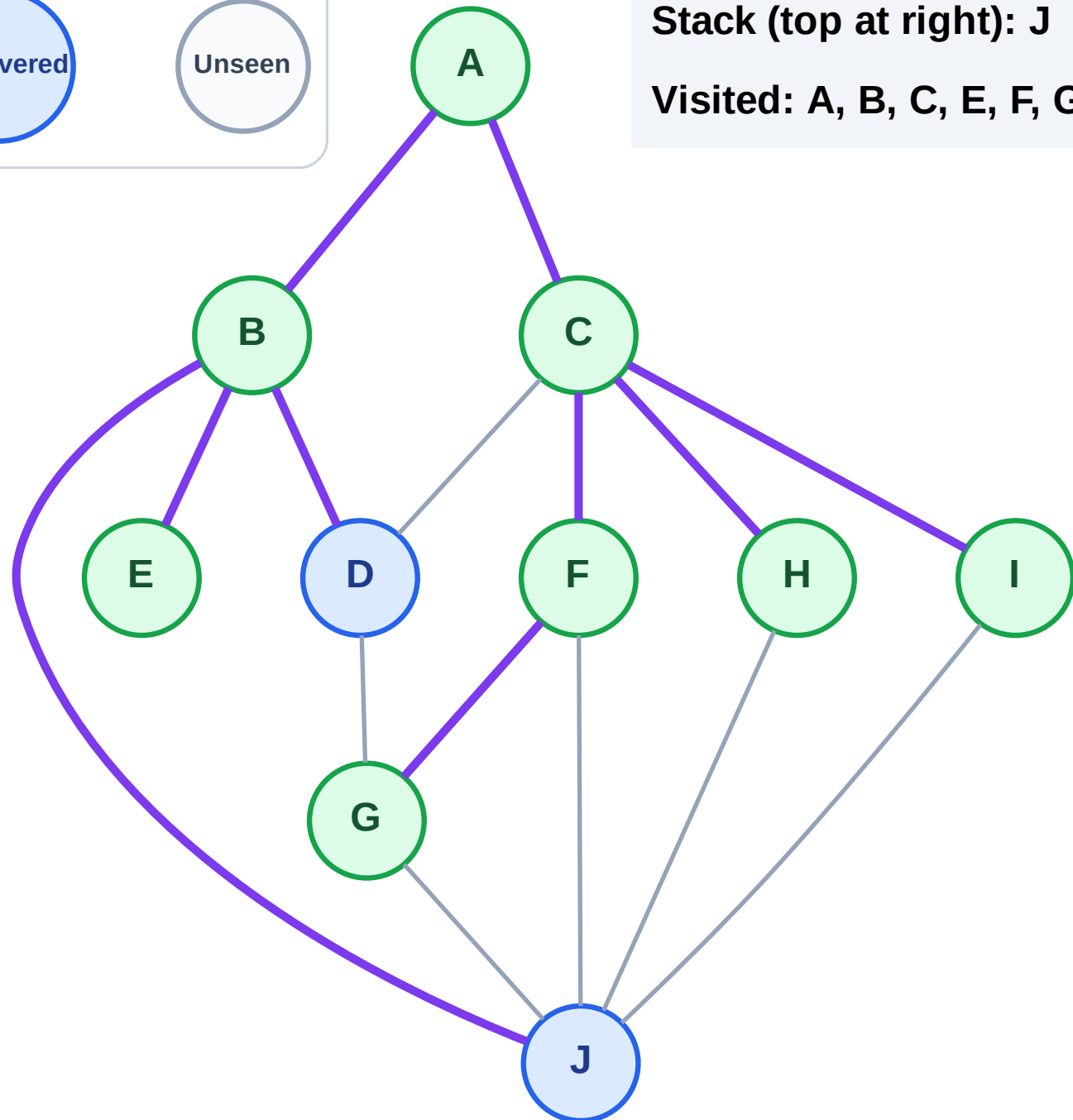
Processing

Discovered

Unseen

Stack (top at right): J | D

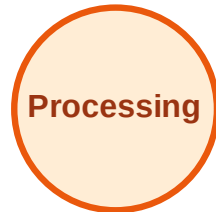
Visited: A, B, C, E, F, G, H, I



DFS Traversal

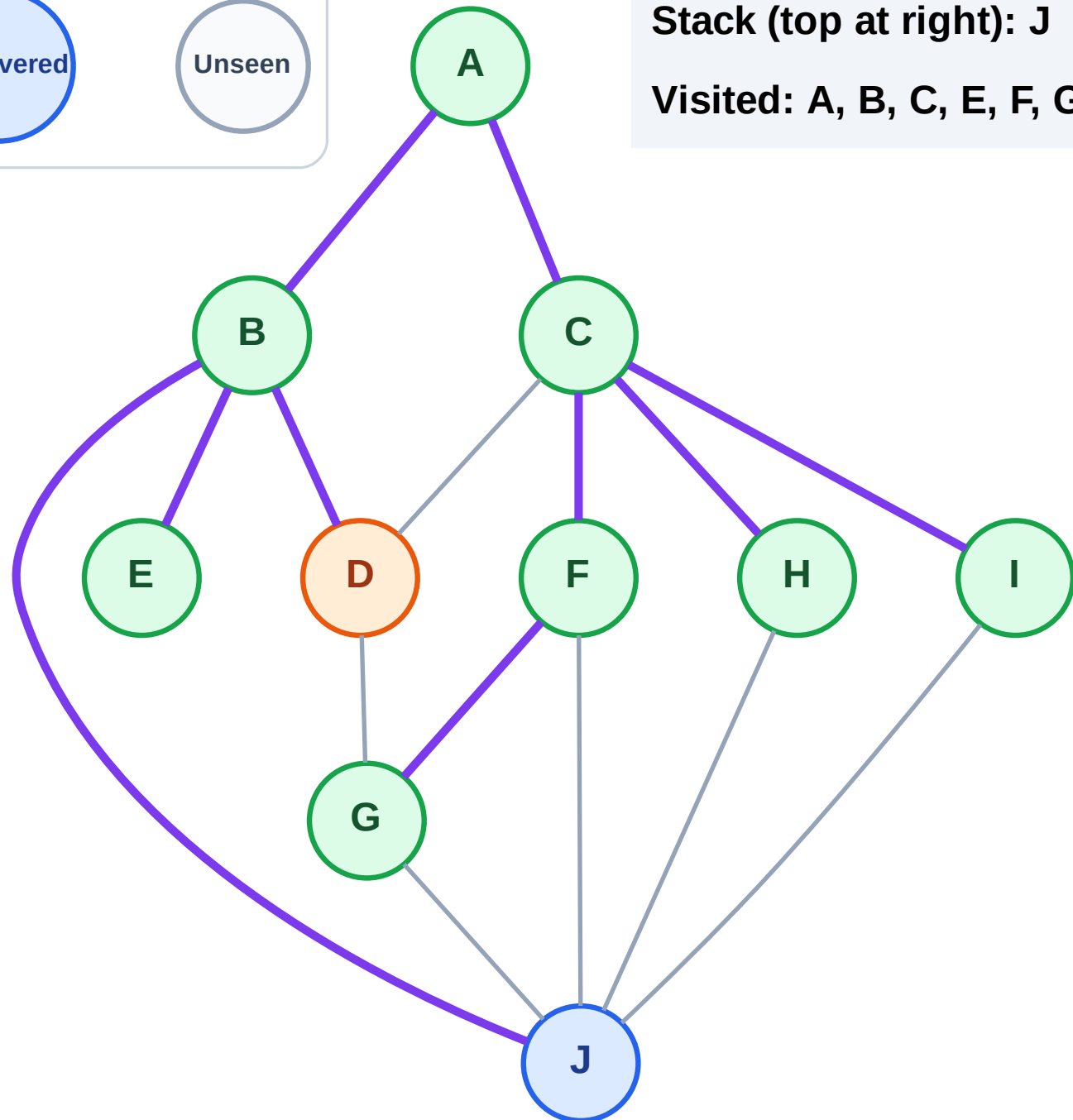
Step 18: Process node D

Legend



Stack (top at right): J

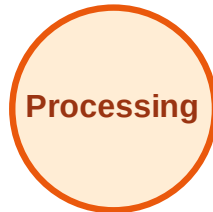
Visited: A, B, C, E, F, G, H, I



DFS Traversal

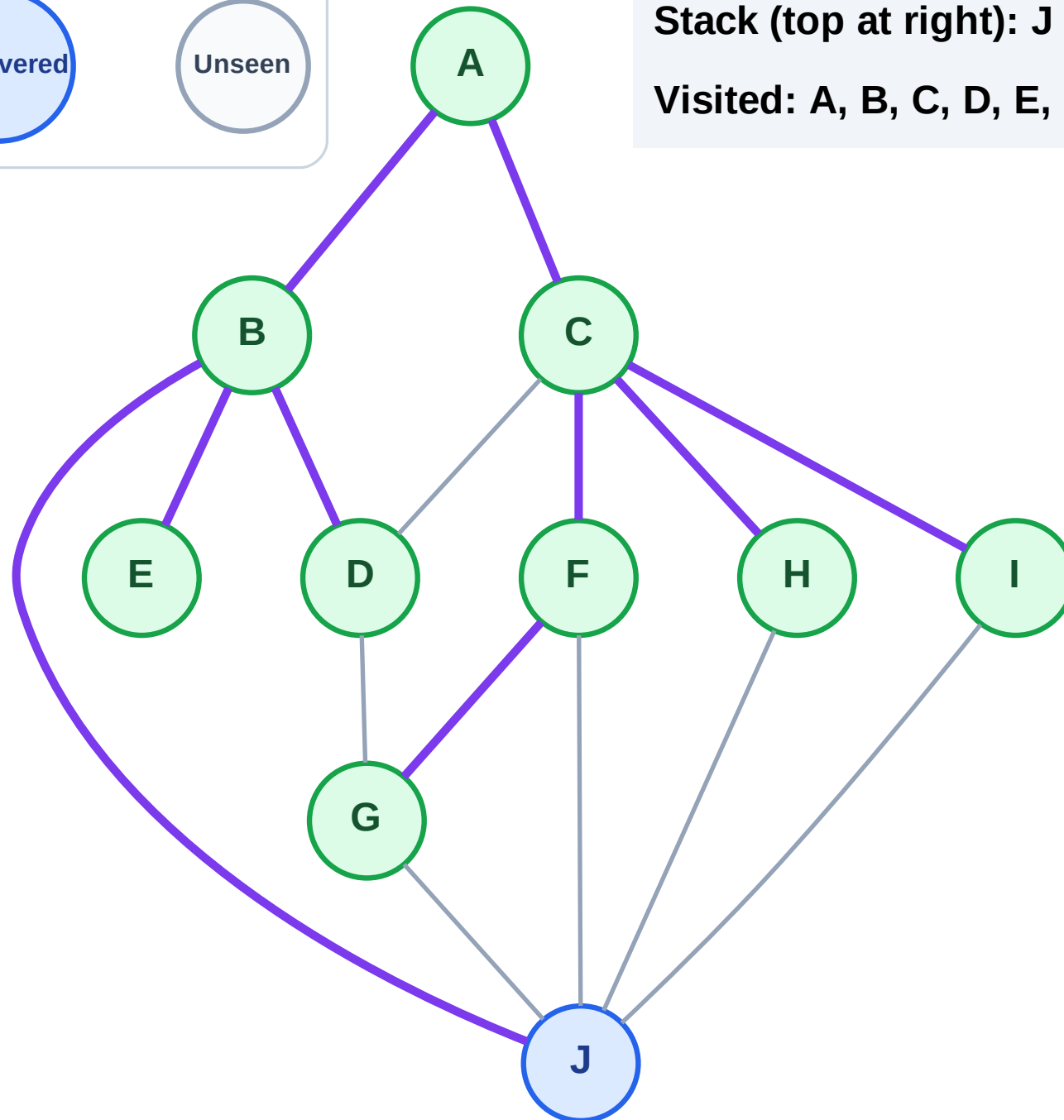
Step 19: No new neighbors from D

Legend



Stack (top at right): J

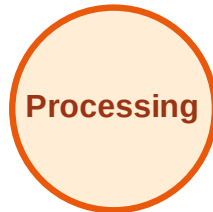
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

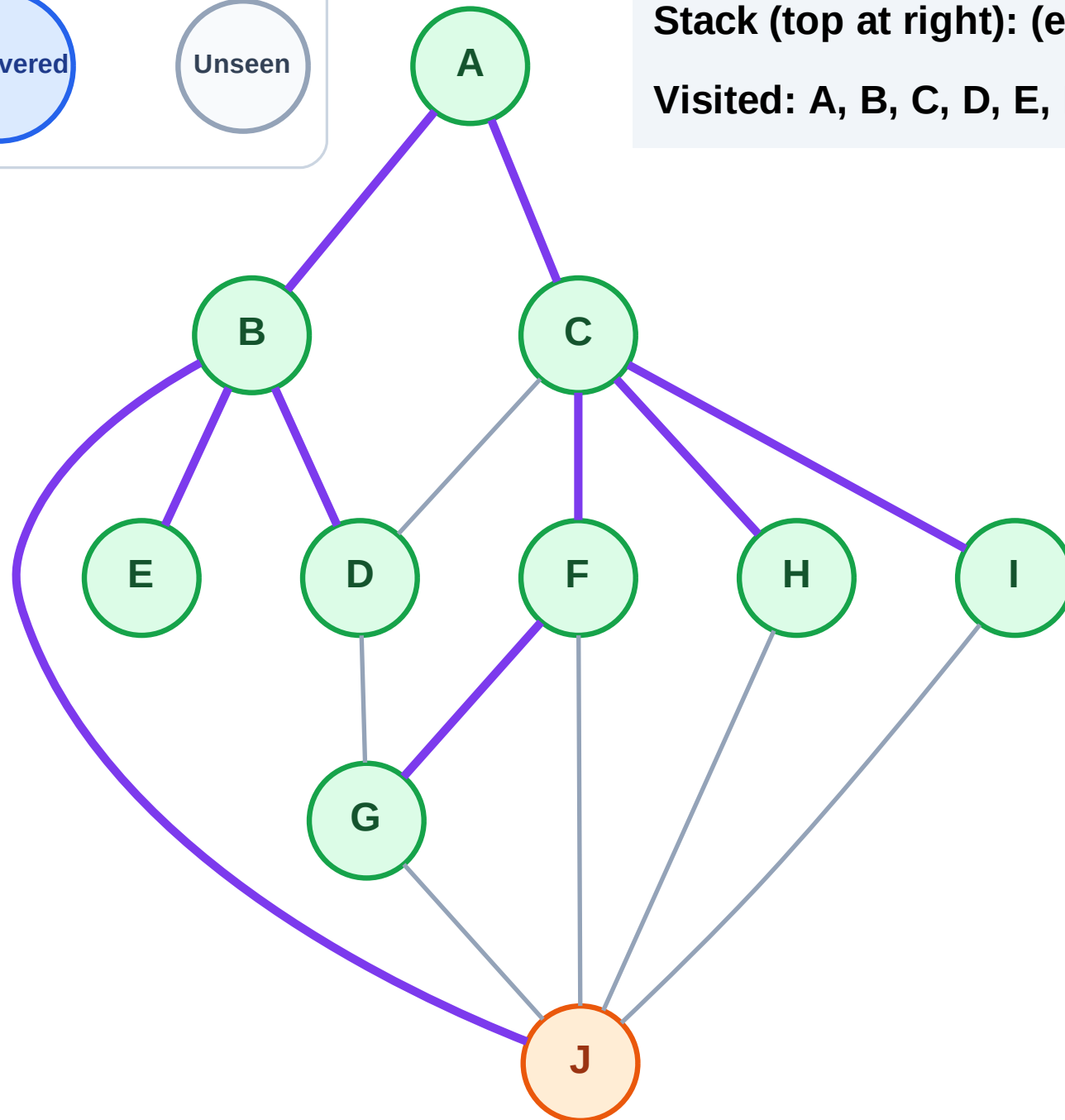
Step 20: Process node J

Legend



Stack (top at right): (empty)

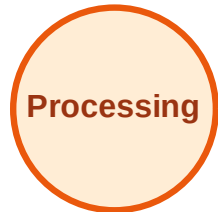
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

